

The Effects of Acquisitions by Digital Platforms on Innovation

Valentina Zvereva[†]

July 2025

Abstract

The study examines the effects of acquisitions by digital platforms on innovation activity. As a proxy for innovation, the number of patents granted to a company is used, reflecting the processes of registration and development of patented technologies. The analysis covers acquisitions by digital platforms of companies operating in the Russian market, including startups. Causal effects of acquisitions on innovation activity are further evaluated using the Dynamic DiD model [Imai et al., 2023]. The results reveal a U-shaped relationship between the average age of patents in an industry and innovation activity. The study finds a short-term negative effect of digital platform acquisitions on innovation two months after the transaction, while no such effect is observed for acquisitions of startups. At the same time, acquisitions by DES have a positive impact after 11 months, whereas acquisitions by digital platforms excluding DES show a short-term positive effect. The paper also discusses potential mechanisms underlying these effects.

Keywords: digital platforms, economic concentration, innovation, digital ecosystems, antitrust regulation

JEL Codes: L41, O31, O32, G34

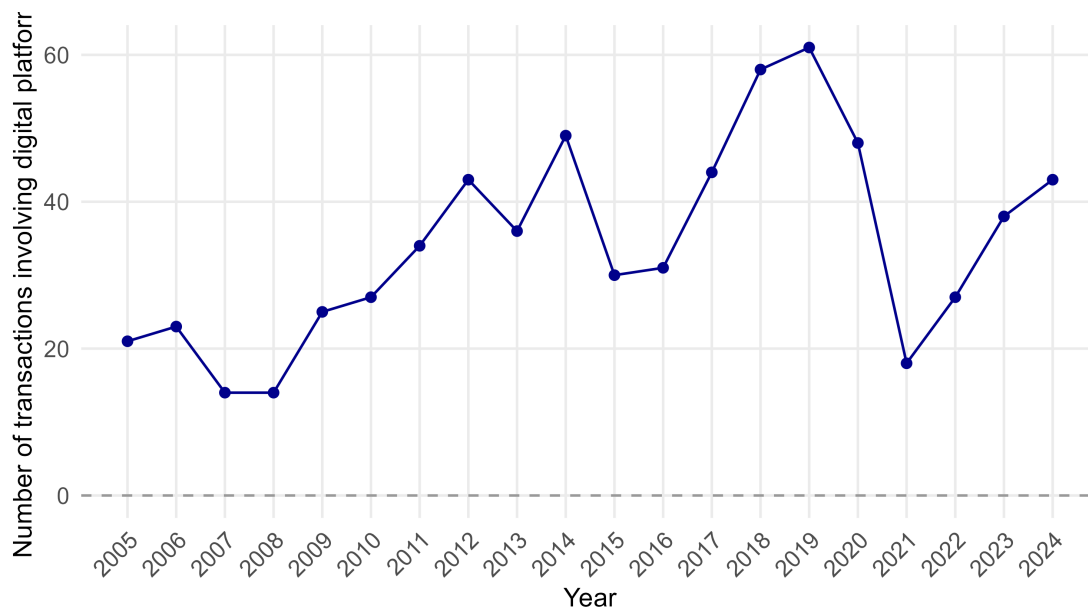
[†]University of Chicago.

Introduction

One of the major transformations of the modern economy over the past 30 years has been digitalization. With the emergence of digital giants such as Google, Yandex, Amazon, and Microsoft, an increasing share of economic activity has moved online. Digital platforms have a substantial impact on the functioning of entire industries. They reshape market structures and the ways market participants interact, while also contributing to economic growth, innovation, and the expansion of data processing. Their importance has become even greater with the emergence of entire ecosystems built around digital platforms [Zvereva Valentina, Research Project e301, 2024].

Among the many challenges that the digital economy poses for competition (including strong network effects, data-driven economies of scale, concerns about personal-data leakage, and others), the economic literature devotes particular attention to the role of economic concentration transactions involving digital platforms. Since 2021, Russia has experienced an increase in the number of economic concentration transactions involving digital platforms¹ (Figure 1).

Figure 1. Number of economic concentration transactions involving digital platforms in Russia, by year.



Source: author's calculations based on Mergers data.

This trend may affect the innovative activity of the merging firms, because acquiring a company also gives a digital platform access to the target's patented technologies, whose future development depends on the platform's strategic objectives for the acquisition. A platform may acquire a startup to strengthen its market power or to obtain patented technologies for further development. Russia also has antitrust exemptions that limit the application of competition law to companies holding exclusive rights to intellectual property.² Digital platforms may therefore have an incentive to acquire companies in order to obtain their patents and use them to create barriers to entry while benefiting from such antitrust exemptions. Moreover, acquisitions may strengthen the market power of digital platforms, an issue that has attracted increasing attention from competition authorities in recent years. Recent amendments to Russian competition law now require merger review for transactions exceeding seven billion rubles.³ This additional criterion is intended to bring within merger control acquisitions by digital platforms that previously did not have to be notified to the competition authority.⁴ Economic theory predicts that, all else equal, a reduction in competitive pressure leads to higher prices, lower output, and lower consumer surplus. At the same time, mergers and acquisitions can affect innovation, potentially lowering prices and improving product quality. In practice,

¹ In this study, digital platforms are understood to mean **transactional digital platforms**, i.e., companies that connect two or more distinct groups of users who join the platform in order to interact directly with one another [Hagiu, Wright, 2015]. A digital platform serves as the foundation for the development of a digital ecosystem. "**Digital ecosystems (DEs)** are a set of economic entities closely connected to a key firm through a digital platform or digital infrastructure and interacting with that firm and with one another via a hybrid transaction-governance mechanism" [Shastitko, Kurdin, Filippova, 2023].

² Part 4 of Article 10 and Part 9 of Article 11 of Federal Law No. 135-FZ "On Protection of Competition."

³ Article 28 of Federal Law No. 135-FZ of July 26, 2006, "On Protection of Competition."

⁴ Startups are particularly relevant in this context because many of them did not meet the previous thresholds based on the book value of assets and revenue, while the transaction value of startup acquisitions is often high. This motivated the introduction of an additional criterion for the approval of economic concentration transactions in order to bring such deals within the scope of merger control [Stavniychuk, Markova, 2023; Markova, Stavniychuk, 2024].

European and U.S. competition authorities take possible positive effects of mergers on innovation into account when assessing merger efficiencies [Morzenti G., 2023]. Existing studies estimate the effect of economic concentration transactions involving digital platforms on innovative activity in the United States and Europe, but comparable estimates for Russia are not available.

The objective of this study is to estimate the effect of economic concentration transactions involving digital platforms, including digital ecosystems, on the innovative activity of Russian firms. The number of patented technologies is used as a proxy for innovative activity, capturing their registration and development.

This paper contributes to the literature in two ways. First, it expands the empirical evidence by providing new estimates for Russia, where studies of this kind are currently absent. Second, it compares transactions involving digital ecosystems with transactions involving other digital platforms, making it possible to identify mechanisms through which such transactions affect innovative activity. The results have practical implications for the design of competition policy and the assessment of transactions involving digital platforms.

1 Effects of Economic Concentration Transactions Involving Digital Platforms and Their Regulation

Mergers and acquisitions (M&A) play a central role in shaping market structure and firms' innovative activity. However, whether M&A promotes or hinders technological progress remains one of the most debated issues in corporate finance and the economics of innovation.

On the one hand, M&A can stimulate innovation by combining knowledge, technologies, and human capital [Ahuja, Katila, 2001]⁵. Post-merger centralization may reduce the transaction costs of coordination and facilitate the development of general-purpose or cross-cutting technologies [Argyres, Silverman, 2004]⁶, thereby generating broader effects on innovation.

M&A may also create complementarities. Large firms with established sales channels but stagnant internal R&D can substantially improve their competitiveness by acquiring firms with strong innovation portfolios, because the target's technologies gain access to the acquirer's resources [Bena, Li, 2014]⁷. Such acquisitions may also stimulate innovation in developing countries: foreign acquirers can provide access to global markets and thereby encourage innovation in acquired subsidiaries [Guadalupe, Kuzmina, Thomas, 2012]⁸. Technological overlap between firms' patent portfolios affects target selection in M&A and may facilitate the creation of complementary assets [Bena and Li, 2014]⁹, which can improve long-run real outcomes and operating performance.

Institutional and cultural differences between acquirers and targets may also matter. Cultural distance between firms can strengthen short-run innovation effects, although these effects weaken after two years [Cloudt, Hagedoorn, Kranenburg, 2006]¹⁰. One possible explanation is that cultural differences facilitate the exchange of resources and ideas and thereby stimulate innovation, although moderate distance may be beneficial while excessive distance may impede integration [Bauer, Matzler, Wolf, 2016]¹¹.

Active M&A markets may also create incentives for small firms: the prospect of being acquired can encourage innovative activity among small companies [Phillips, Zhdanov, 2013]¹². At the same time, empirical studies suggest that the positive effect of startup acquisitions by large digital platforms is temporary: approximately 1.5 years after an acquisition, the effect is no longer statistically significant [de Barys, Gautier, 2024]¹³.

Acquisitions of digital firms, particularly in cross-border transactions, often have more positive effects on innovation because digital technologies can be applied directly in the innovation process [Hanelt et

⁵Ahuja, G., & Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3), 197–220.

⁶Argyres, N. S., & Silverman, B. S. (2004). R&D, organization structure, and the development of corporate technological knowledge. *Strategic Management Journal*, 25(8–9), 929–958.

⁷Bena, J., & Li, K. (2014). "Corporate Innovations and Mergers and Acquisitions." *The Journal of Finance*, Vol. 69, No. 1, pp. 201–231.

⁸Guadalupe, M., Kuzmina, O., & Thomas, C. (2012). Innovation and foreign ownership. *American Economic Review*, 102(7), 3594–3627.

⁹Bena, J., & Li, K. (2014). "Corporate Innovations and Mergers and Acquisitions." *The Journal of Finance*, Vol. 69, No. 1, pp. 201–231.

¹⁰Cloudt, M., Hagedoorn, J., & Van Kranenburg, H. (2006). Mergers and acquisitions: Their effect on the innovative performance of companies in high-tech industries. *Research Policy*, 35(5), 642–654.

¹¹Bauer, F., Matzler, K., & Wolf, S. (2016). M&A and innovation: The role of integration and cultural differences—A central European targets perspective. *International Business Review*, 25(1), 76–86.

¹²Phillips, G. M., & Zhdanov, A. (2013). R&D and the incentives from merger and acquisition activity. *The Review of Financial Studies*, 26(1), 34–78.

¹³Barys L., Gautier A. Big Tech Acquisitions and Innovation: An Empirical Assessment // CESifo Working Paper No. 11025. – 2024.

al., 2021¹⁴; Xu, Deng, 2024¹⁵. An additional positive mechanism is that foreign acquirers often relax the financial constraints faced by acquired firms in developing countries, enabling them to invest more actively in R&D and improve productivity [Chen, Hua, Boateng, 2017¹⁶].

The consequences of M&A depend on the combination of the startup's resources and the acquirer's incentives. [Motta, Peitz, 2021¹⁷] show that an acquisition may be neutral, efficiency-enhancing, anticompetitive, or constitute a "killer acquisition," in which a potential innovation is deliberately suppressed. This ambiguity may arise because a merger can simultaneously promote innovation through coordination and discourage it through the internalization of externalities—in particular, through cannibalization of the firm's existing product by a new product, which reduces incentives to develop the latter [Federico, Langus, Valletti, 2018¹⁸]. Effects may also differ depending on the mechanism used to reallocate resources within the merged firm and on changes in the direction of innovation [Moraga-González et al., 2022¹⁹; Dijk, Moraga-González, Motchenkova, 2024²⁰].

These mechanisms also point to potential negative effects of M&A. Such transactions may alter firms' incentives by strengthening their focus on short-run results and reducing the quality of patents they produce [Valentini, 2012²¹]. Patent output and R&D may decline both for the merged firm and for its competitors, especially in R&D-intensive markets [Szücs, 2014²²; Haucap, Rasch, Stiebale, 2019²³; Rogova, Kirichenko, Lebedeva, 2023²⁴; Gugler et al., 2025²⁵]. Moreover, the post-M&A decline in innovative activity is larger among firms with higher-quality patents and in more concentrated markets [Altomonte C. et al., 2024²⁶].

One explanation for the negative effect of M&A on innovation concerns the distributional consequences of these transactions and the strategic motives of acquirers. In particular, firms may engage in so-called killer acquisitions [Ornaghi, 2009²⁷; Haucap, Rasch, Stiebale, 2019²⁸], which reduce the activity of projects that overlap with the acquirer's patent portfolio, especially when market power is high [Cunningham, Ederer, Ma, 2021²⁹].

Observed effects of M&A on patenting are heterogeneous. One reason may be firm selection: acquired companies are initially more innovative and spend more on R&D, but reduce innovation investment after the transaction [Szücs, 2014³⁰]. Heterogeneity in outcomes may also be amplified by information asymmetry between the buyer and the seller (target): after M&A transactions with more transparent terms, firms exhibit better financial and operating performance [Thompson, Kim, 2020³¹].

Negative effects of M&A in innovative industries may also operate through acqui-hiring: a startup is acquired primarily as a source of human capital, particularly in the IT industry. As a result, employees of the target firm may fail to realize their innovative potential within the acquiring company [Benkert,

¹⁴Hanelt, A., Firk, S., Hildebrandt, B., & Kolbe, L. M. (2021). Digital M&A, digital innovation, and firm performance: an empirical investigation. *European Journal of Information Systems*, 30(1), 3–26.

¹⁵Xu, H., & Deng, S. (2024). Digital Mergers and Acquisitions and Enterprise Innovation Quality: Analysis Based on Research and Development Investment and Overseas Subsidiaries. *Sustainability*, 16(3), 1120.

¹⁶Chen, Y., Hua, X., & Boateng, A. (2017). Effects of foreign acquisitions on financial constraints, productivity and investment in R&D of target firms in China. *International Business Review*, 26(4), 640–651.

¹⁷Motta M., Peitz M. Big tech mergers. // *Information Economics and Policy*. – 2021. – Vol. 54.

¹⁸Federico G., Langus G., Valletti T. Horizontal mergers and product innovation // *International Journal of Industrial Organization*. – 2018. – Vol. 59. – P. 1–23.

¹⁹Moraga-González, J. L., Motchenkova, E., and Nevrekar, S. Mergers and innovation portfolios // *The RAND Journal of Economics*. – 2022. – Vol. 53. – P. 641–677.

²⁰Dijk E., Moraga-González J., Motchenkova E. How Do Start-up Acquisitions Affect the Direction of Innovation? // *The Journal of Industrial Economics*. – 2024. – Vol. 72. - Issue 1. – P. 118–156.

²¹Valentini, G. Measuring the effect of M&A on patenting quantity and quality // *Strategic Management Journal*. – 2012. – Vol. 33. – Issue 3. – P. 336–346.

²²Szücs F. M&A and R&D: Asymmetric Effects on acquirers and targets? // *Research Policy*. – 2014. – Vol. 43. – P. 1264–1273.

²³Haucap J., Rasch A., Stiebale J. How mergers affect innovation: Theory and evidence // *International Journal of Industrial Organization*. – 2019.- Vol. 63 – P. 283–325.

²⁴Rogova, E., Kirichenko, M., & Lebedeva, D. (2023). Do Mergers and Acquisitions Promote R&D? The Case of European Innovation and Technology Sector. *Corporate Finance*, 17(4), 5–18.

²⁵Gugler K., Szücs F., Wohak U. Start-up acquisitions, venture capital and innovation: A comparative study of Google, Apple, Facebook, Amazon and Microsoft // *International Journal of Industrial Organization*. – 2025. – Vol. 99.

²⁶Altomonte C., El-Mallakh N., Sonno T. Business groups, strategic acquisitions and innovation // *CEP Discussion Papers No. 1996*. – 2024.

²⁷Ornaghi, C. (2009). Mergers and innovation in big pharma. *International Journal of Industrial Organization*, 27(1), 70–79.

²⁸Haucap J., Rasch A., Stiebale J. How mergers affect innovation: Theory and evidence // *International Journal of Industrial Organization*. – 2019.- Vol. 63 – P. 283–325.

²⁹Cunningham, C., Ederer, F., and Ma, S. Killer acquisitions // *Journal of Political Economy*. – 2021. – Vol. 129. – No. 3. – P. 649–702.

³⁰Szücs F. M&A and R&D: Asymmetric Effects on acquirers and targets? // *Research Policy*. – 2014. – Vol. 43. – P. 1264–1273.

³¹Thompson, E. K., & Kim, C. (2020). Information asymmetry, time until deal completion and post-M&A performance. *Journal of Derivatives and Quantitative Studies: 선물연구*, 28(3), 123–140.

Letina, Liu, 2025³²]. In addition, acqui-hiring may increase employers' labor-market power and reduce wages [Bar-Isaac, Johnson, Nocke, 2025]³³.

Innovation quality is usually assessed through a patent's impact on subsequent inventions, generality of application, and originality [Argyres and Silverman, 2004; Hall, Jaffe, Trajtenberg, 2001]. Empirical studies based on data from the U.S. medical-device and photographic-equipment industries (1988–1996) show that M&A increases the number of patents but reduces their quality. This evidence is consistent with the hypothesis that transactions may redistribute knowledge without substantially increasing its marginal innovative value.

Later studies extend these findings by showing that the innovation effects depend on the type of transaction and the structure of the industry. “How Mergers Affect Innovation: Theory and Evidence” [Haucap, Rasch, and Stiebale, 2019], using propensity score matching and difference-in-differences (DiD), finds that patenting and R&D decline both at the merged firm and among its competitors, especially in R&D-intensive markets. Similar conclusions are reported in “M&A and R&D: Asymmetric Effects on Acquirers and Targets?” [Szücs, 2014] and “M&A and Innovation in Big Tech” [Gugler K. et al., 2025], where approximately 14% of acquisitions have a negative effect on both innovation and investment.

At the same time, the effect of M&A may vary with the characteristics of the parties to the transaction. “Business Groups, Strategic Acquisitions and Innovation” [Altomonte C. et al., 2024], which uses TWFE Difference-in-Differences and patent-portfolio similarity measures, finds a decline in innovative activity after acquisitions, especially among firms with high-quality patents and in highly concentrated markets. By contrast, cultural distance between firms may strengthen short-run innovation effects, although these effects weaken after two years [Clodt M. et al., 2006].

Recent empirical work increasingly focuses on startup acquisitions by large digital platforms. “Big Tech Acquisitions and Innovation: An Empirical Assessment” [Laureen de Barsey and Axel Gautier, 2024] uses a sharp event-study design and a control group of comparable patents to show that the positive effect of such transactions is temporary: it loses statistical significance approximately 1.5 years after the acquisition. In the pharmaceutical industry, so-called killer acquisitions have been documented [Cunningham C., Ederer F., Ma S., 2021]; these transactions reduce activity in projects that overlap with the acquirer's patent portfolio, especially when market power is high.

Empirical studies of these effects typically use panel data across firms, industries, and time (years or months), with M&A transactions occurring asynchronously. This setting requires methods that account for temporal and group heterogeneity as well as violations of the standard DiD assumptions. Common approaches include event studies, negative binomial regressions, propensity score matching (PSM), DiD, triple differences, staggered-adoption DiD, and TWFE. These methods can help address the endogeneity of transactions and control for factors such as overlap in patent portfolios or industry structure.

The literature also pays particular attention to the effect of antitrust regulation on M&A and innovation. In 2000, the United States amended the Hart–Scott–Rodino Act by raising the transaction-value threshold for mandatory merger notification. This reduced the number of merger notifications by 70% and created scope for “stealth consolidation”: small transactions escaped review even though their cumulative effect was substantial. “Stealth Consolidation: Evidence from an Amendment to the Hart–Scott–Rodino Act” [Thomas G. Wollmann, 2019] uses Difference-in-Differences and Triple Difference-in-Differences and shows that weaker review increased the number of transactions between competitors and thereby increased market concentration.

Weaker antitrust enforcement may also affect innovative activity. “Antitrust Policy and Innovation” [Morzenti G., 2023] exploits a natural experiment. The treatment group consists of firms involved in horizontal mergers exempted from regulation, while the control group consists of firms involved in vertical transactions and firms not subject to the regulatory relaxation. The effect is estimated using Triple Difference-in-Differences and an NLP-based transaction classification. The results suggest that weaker regulation reduces patenting by 30%, and leave-one-out exercises support the robustness of the findings.

The long-run dynamic consequences of antitrust policy are studied in general-equilibrium models. “The Dynamic Effects of Antitrust Policy on Growth and Welfare” [Cavenaile L. et al., 2021] shows that stronger merger control increases innovation by “superstar” firms but reduces R&D by small firms because it lowers their expectations of future M&A opportunities.

Overall, both theoretical and empirical studies show that the effect of M&A on innovation depends on a combination of factors, including market structure, the strength of regulation, and the type of firms involved. Transactions may stimulate innovation in the short run, while their cumulative effect under weaker enforcement may contribute to market concentration and lower aggregate innovative activity. Dynamic DiD, event-study, and matching methods make it possible to account for this heterogeneity and identify both positive and negative effects of M&A. This study contributes to the literature by providing new evidence on the dynamic effects of transactions in the Russian market and by showing that the effect

³²Benkert, J. M., Letina, I., & Liu, S. (2025). Startup acquisitions: Acqui-hires and talent hoarding. *European Economic Review*, 105103.

³³Bar-Isaac, H., Johnson, J. P., & Nocke, V. (2025). Acqui-hiring for monopsony power. *Management Science*, 71(4), 3485–3496.

of M&A on innovation depends on the type of acquirer, time since the transaction, and industry maturity.

2 Hypotheses

As discussed above, the literature does not provide an unambiguous answer regarding the direction of the effect of mergers and acquisitions involving digital platforms on innovative activity. Because digital platforms in the Russian market tend to acquire firms operating in the same industries as the platforms themselves (services and trade, telecommunications), the **hypotheses** of this study are as follows:

- 1) Economic concentration transactions involving digital platforms have a negative effect on the innovative activity of the merging firms;
- 2) There is a U-shaped relationship between the number of economic concentration transactions involving digital platforms and innovative activity.

3 Empirical Assessment of the Effects of Economic Concentration Transactions Involving Digital Platforms on Innovative Activity

3.1 Data and Preliminary Analysis

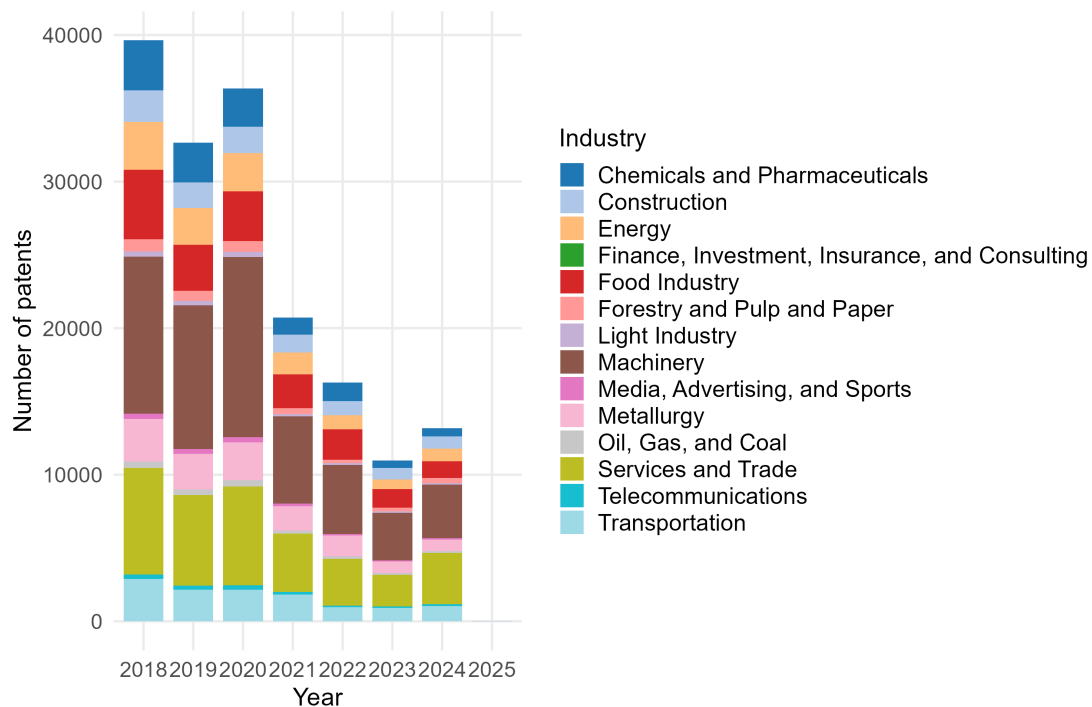
3.1.1 Patent Data

To construct the patent dataset, I use the Yandex Patents service,³⁴ which contains Russian patents for utility models and inventions as well as patent applications. Although other forms of innovation protection exist, such as trade secrecy, industries within the same country operate under the same institutional environment. This makes patents a useful proxy for innovative activity, while preserving the internal validity of cross-industry comparisons. The service contains data on patent applications through 2023. Data for 2024–2025 were collected from the Rospatent search platform.³⁵

Data from Yandex Patents were obtained by scraping the website using the Selenium library, which is designed for interacting with dynamic web pages. Data from the Rospatent platform were collected manually.

The largest number of patents belongs to the machinery industry (Figure 2).

Figure 2. Number of patents by industry and application year, 2018–2025.



Source: author's calculations based on Yandex Patents and Rospatent data.

3.1.2 Data on Economic Concentration Transactions

Data on economic concentration transactions from 2001 to 2025 were collected from the Mergers portal.³⁶ The Mergers database contains information on the parties to each transaction (seller, buyer, and target);

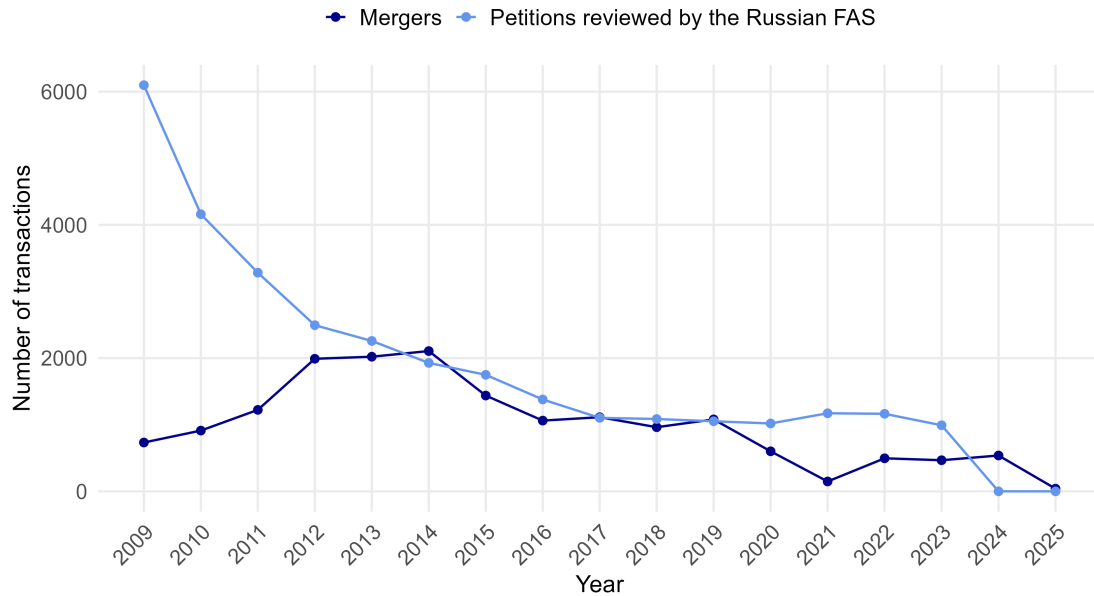
³⁴Yandex Patents — patent-document search

³⁵Rospatent search platform

³⁶Mergers and Acquisitions in Russia / Mergers.ru

characteristics of the participants (target, industry, region, address, website, and description) and of the transaction (status, price, stake, category, type, direction, and nationality); dates (record creation, update, negotiations, signing, and completion); and related news (transaction news, website news, and external news). Beginning in 2012, the number of transactions recorded in Mergers is close to the number of economic-concentration petitions reviewed by the Federal Antimonopoly Service (FAS), suggesting that the database is sufficiently complete and relevant for the subsequent analysis (Figure 3).

Figure 3. Number of unique transactions in the Mergers database and economic-concentration petitions reviewed by the Russian FAS, 2009–2025.



Source: author's calculations based on Mergers and Russian FAS data.

Following [Stavniychuk A.Yu., Markova O.A., 2024], I use the preview date as the transaction date because this variable has the fewest missing values relative to the other date variables across industries. Transactions may have one of several statuses: “completed,” “documentation,” “negotiations,” or “cancelled.” The database contains data-entry errors in these fields: some transactions have a recorded completion date but are not assigned the status “completed.” Thus, $(6,678 + 205) / 29,317 = 23.5\%$ of transactions are classified inconsistently, while $(16,692 + 5,742) / 29,317 = 76.5\%$ are classified consistently (Table 1).

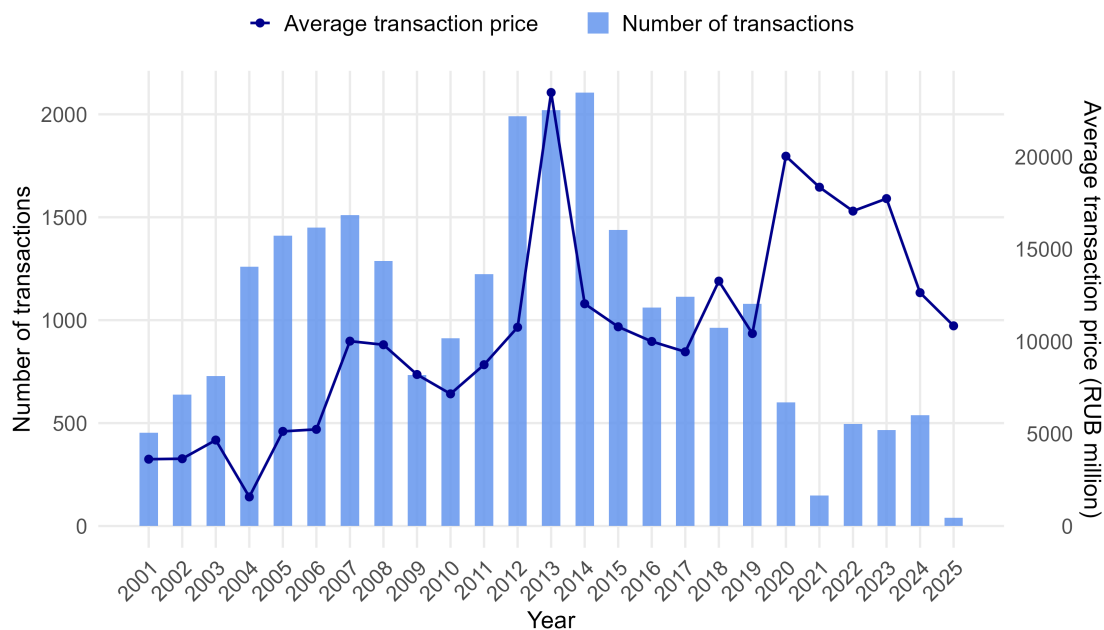
Table 1: Number of Deals by Completion Status and Availability of a Completion Date

		Deal Status		Row Total
		Completed	Not Completed	
Completion Date	Available	16,692	205	16,897
	Not Available	6,678	5,742	12,420
Column Total		23,370	5,947	29,317

Source: Author's calculations based on Mergers data.

The analysis focuses on completed transactions. Given the data issue described above, I classify a transaction as completed if either its status is “completed” or a completion date is recorded. For these transactions, I calculate the average transaction price by year and the number of transactions by year (Figure 4).

Figure 4. Average transaction price (RUB million) and number of transactions by year, 2001–2023.

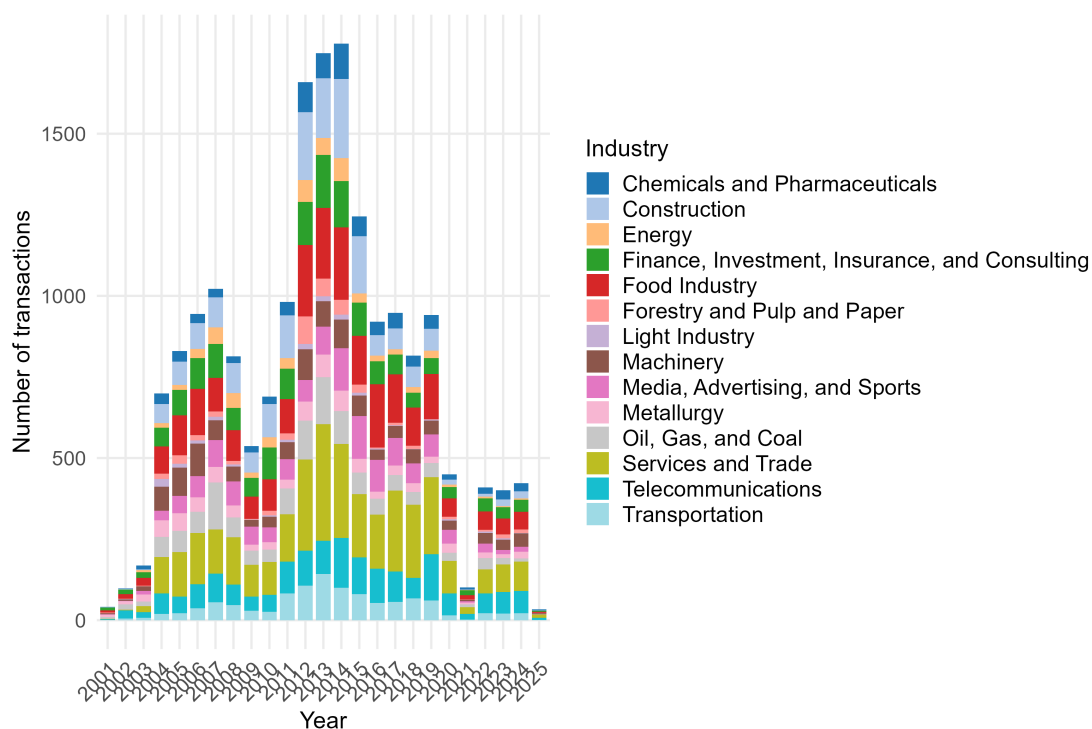


Source: author's calculations based on Mergers data.

Historically, firms involved in mergers and acquisitions were most often from the food industry. However, the share of food-industry firms declined between 2012 and 2024, and in recent years telecommunications firms have accounted for the largest share of transactions (Figure 5).

To estimate the effect of economic concentration transactions between digital platforms and startups, I use the Rusbase database,³⁷ which contains data on startup acquisitions and investments from January 2010 through April 2023.

Figure 5. Number of M&A transactions by industry and year, 2001–2025.



Source: author's calculations based on Mergers data.

³⁷Market Deals Encyclopedia | RB.DATA

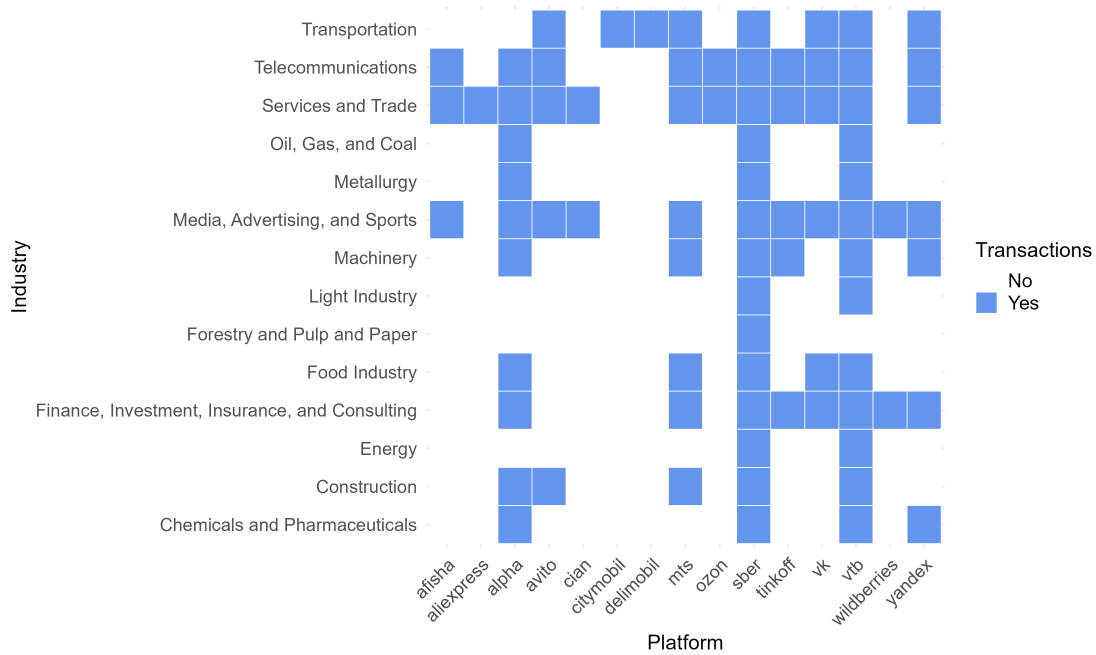
Using information on the acquirer, I construct the variable

$$treatment_platform = \begin{cases} 1, & \text{if the acquirer in the transaction was a digital platform;} \\ 0, & \text{otherwise.} \end{cases}$$

The set of digital platforms used in the subsequent analysis is based on the definition of transaction platforms in [Hagiu, Wright, 2015]. Under this definition, labor-market platforms (e.g., HeadHunter) and information or reference services (Gosuslugi, 2GIS) are not included in the treatment group. The digital-platform sample includes VK (Mail.ru Group), Yandex, SberBank, MTS, T-Bank (Tinkoff), Ozon, Avito, Wildberries, Cian, Afisha, Delimobil, and AliExpress. Experts from Skolkovo identify VK, Sber, Yandex, and MTS as the largest digital ecosystems.³⁸ The subsequent analysis of economic concentration transactions involving digital ecosystems therefore uses data for these companies.

The industry distribution of M&A transactions involving platforms is uneven. Sber completed transactions with firms in every industry in the sample except light industry, whereas Citymobil and Delimobil completed transactions only in the industry corresponding to their core business (Figure 6).

Figure 6. Distribution of digital-platform transactions across Mergers industries.



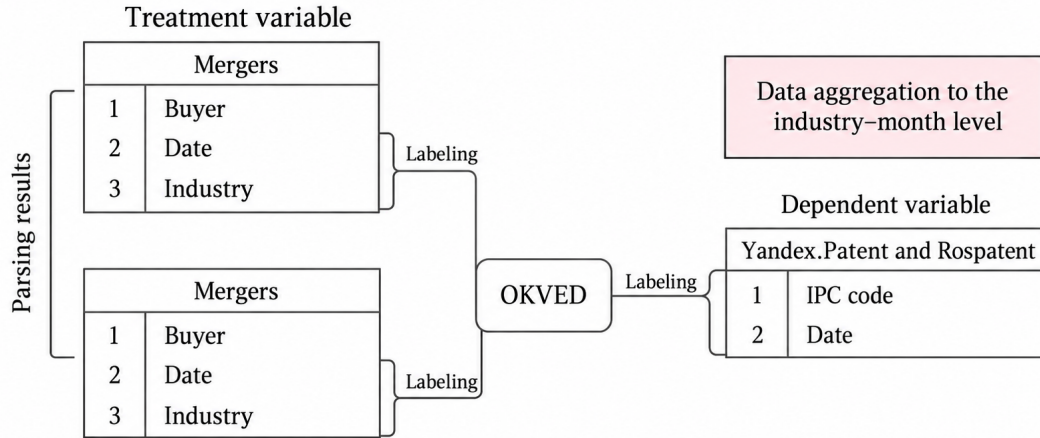
Source: author's calculations based on Mergers data.

3.1.3 Aggregating Patent-Activity and M&A Transaction Data

I next aggregate patent data, Rusbase transactions, and Mergers transactions to the industry-by-month level. Because the data do not contain firms' taxpayer identification numbers (INNs), it is not possible to match firm-level transaction records to patent records and therefore to conduct the analysis at the firm level. Moving to the industry level requires identifying the industry in which each patent is used. Since the patent data do not contain this information, I use International Patent Classification (IPC) codes to assign patents to industries. I map IPC codes in the patent data to OKVED industry-classification codes and then use the OKVED codes to harmonize industries across the Rusbase and Mergers datasets, which use different industry labels. The dependent variable constructed from the patent data is the monthly number of patents in each industry. Treatment variables are constructed from the Mergers and Rusbase data. I separately identify digital platforms and digital ecosystems. In the subsequent analysis, treatment is defined as the occurrence in an industry-month of a transaction involving a digital platform, a digital ecosystem, or a digital platform excluding digital ecosystems (Figure 7).

Figure 7. Construction of the dependent variable and treatment variable.

³⁸RBC. Experts identify companies exhibiting features of ecosystems. URL: https://www.rbc.ru/technology_and_media/01/02/2022/61f3d76f9a794775ff544309 [accessed April 7, 2025].



Source: constructed by the author.

Source: constructed by the author.

The final dataset containing all Mergers transactions includes 1,440 observations for 14 industries over 2018–2025. The final dataset of startup transactions from Rusbase contains 2,108 observations for 34 industries over 2018–2023. The largest numbers of M&A transactions and transactions involving digital platforms occur in services and trade (Table 2). The largest number of patents is observed in machinery. The only industry in which digital platforms did not complete any transactions between 2018 and 2025 is forestry and pulp and paper. Only one transaction involving a digital platform, Sber, took place in light industry.

Table 2: Number of M&A Deals and Patents by Industry, 2018–2025

Industry	Total Deals	Platform Deals	Patents
Chemicals and Pharmaceuticals	176	5	12,218
Construction	198	5	9,512
Energy	63	3	12,338
Finance, Investment, Insurance, and Consulting	259	30	0
Food Industry	489	13	18,121
Forestry and Pulp and Paper	50	0	3,430
Light Industry	18	1	1,489
Machinery	227	9	50,366
Media, Advertising, and Sports	234	27	1,539
Metallurgy	130	2	12,580
Oil, Gas, and Coal	181	3	1,902
Services and Trade	845	70	32,975
Telecommunications	493	69	1,387
Transportation	209	22	11,962

Notes: The table reports the number of merger and acquisition deals, deals involving digital platforms, and patents by industry from 2018 to 2025.

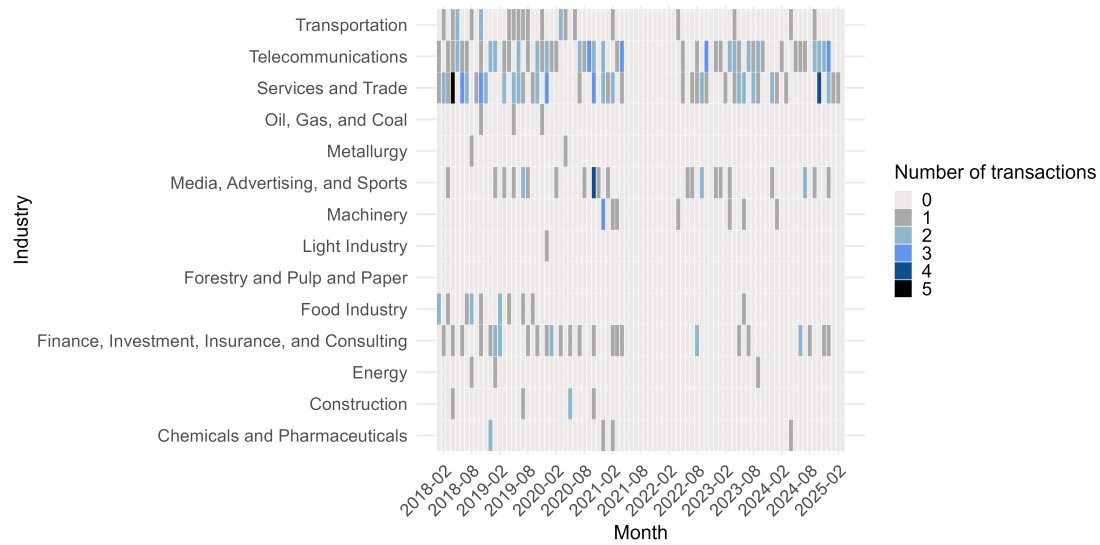
Source: Author's calculations based on Mergers, Rospatent, and Yandex Patents data.

The average age of patents in an industry-month is 914 days. The maximum number of patents in an industry-month is 2,232, while the average number of patents in an industry is 914. The average share of horizontal mergers in an industry-month is 35.7% (Table 3).

Most transactions involving digital platforms occur in services and trade and in telecommunications. In the second half of 2021, digital platforms did not complete transactions with firms in the sample (Figure 8). *Figure 8. Monthly distribution of the number of transactions involving digital platforms across industries, 2018–2025.*

Table 3: Descriptive statistics for data aggregated to the industry-month level

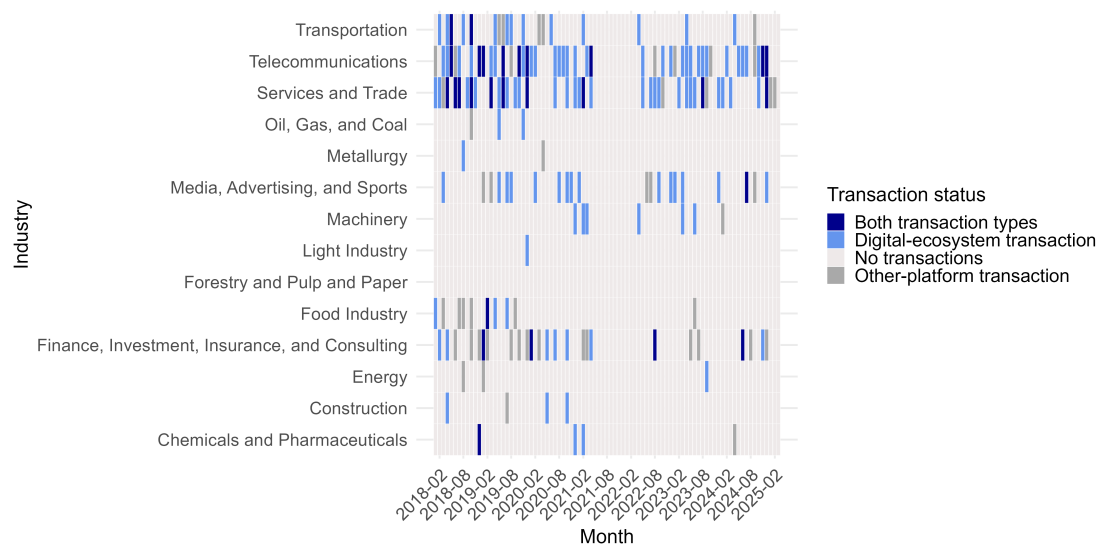
Variable Name	Mean	Standard Deviation	Minimum	Maximum
num_platform_deals	0.213	0.57	0	5
num_deals	2.933	4.064	0	32
capacity	2.728	3.74	0	29
horizontal	0.357	0.387	0	1
volume	256,139,610.016	905,739,624.704	0	18,713,000,000.000
num_patents	139.424	203.93	0	2232
avg_patents_age	913.366	634.701	0	2205.419



Source: author's calculations based on Mergers, Yandex Patents, and Rospatent data.

To examine potential treatment-effect heterogeneity, I distinguish between two types of transactions in the subsequent analysis: transactions involving digital platforms other than digital ecosystems, and transactions involving digital ecosystems. Digital ecosystems mostly acquire firms in industries in which they themselves operate. By contrast, digital platforms excluding digital ecosystems acquire firms more evenly across industries (Figure 9).

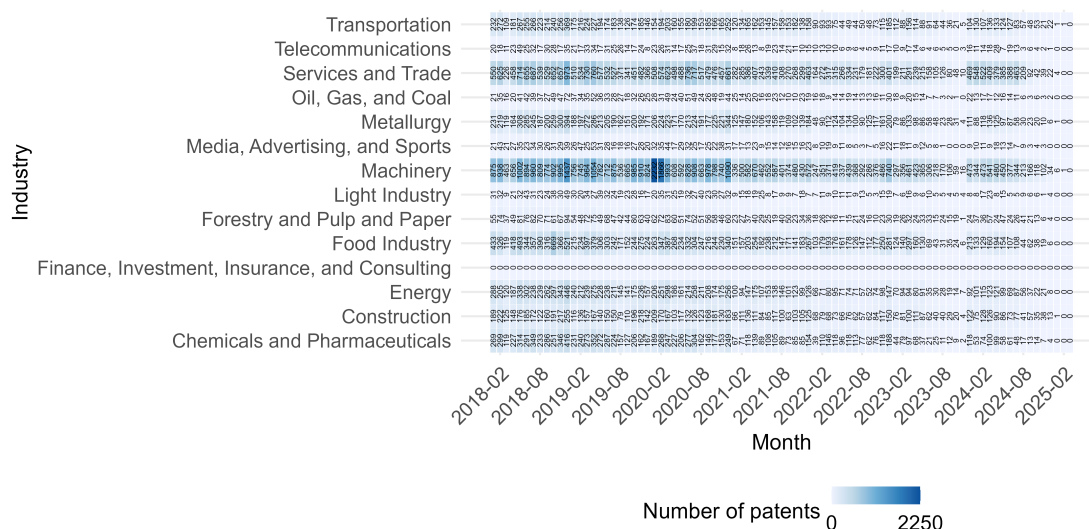
Figure 9. Distribution of transactions involving digital ecosystems and other digital platforms by month and industry, 2018–2025.



Source: author's calculations based on Mergers, Yandex Patents, and Rospatent data.

I next examine the distribution of patents across industries and months. No patents are recorded in the finance industry. Machinery has the largest number of patents, followed by services and trade (Figure 10). Services and trade also has the largest number of economic concentration transactions involving digital platforms. Telecommunications, which has the second-largest number of such transactions, has a maximum of only 49 patents per month, substantially fewer than services and trade. I account for this difference in the estimation by including the cumulative number of patents in the industry among the matching variables.

Figure 10. Distribution of patents across industries and months, 2018–2025.



Source: author's calculations based on Mergers, Yandex Patents, and Rospatent data.

Finally, I examine the distribution of digital-platform transactions with startups. In this subsample, transactions are concentrated primarily in services and software. There are many industries in which digital platforms did not complete any startup transactions. Most startup transactions involve digital ecosystems (Figure 11).

Figure 11. Distribution of startup transactions involving digital ecosystems and other digital platforms by month and industry, 2018–2023.



Source: author's calculations based on Rusbase, Yandex Patents, and Rospatent data.

3.2 Empirical Strategy

3.2.1 Choice of Estimation Method Given the Available Assumptions

As shown in Section 2, the main methods used in the existing literature to estimate the effects of economic concentration transactions are dynamic difference-in-differences, event studies with different forms of matching, staggered-adoption DiD with propensity score matching, and TWFE models used to estimate associations (Table 1). In the analysis that follows, treatment is defined as a transaction involving a digital platform or digital ecosystem in a given industry-month. Treatment in the data is dynamic, because it occurs at different points in time, and recurrent, because the same industry may be treated multiple times, including in consecutive months. The data have a panel structure, and treatment is endogenous. Previous treatment is a confounder because it affects both the dependent variable—earlier transactions involving digital platforms may affect the number of patents with a lag—and current treatment, since an industry in which a digital-platform transaction has already occurred may become more attractive for acquisitions by competing platforms.

A standard DiD design assumes a one-time treatment, which does not correspond to the structure of my data because treatment is dynamic. A staggered-adoption DiD design is also not applicable under this treatment pattern because treatment is recurrent, making it impossible to divide industries into a stable treatment group and a control group. A standard synthetic-control design is likewise not suitable because it assumes a one-time treatment. Moreover, obtaining a robust synthetic-control estimate requires a long pre-treatment period, which is also not characteristic of my data. Generalized synthetic control [Xu, 2017] allows for more than one treatment event, but one of its assumptions is treatment irreversibility; in addition, this approach requires a sufficiently large number of control units that are never treated. To estimate the effect of economic concentration transactions involving digital platforms on innovative activity, I therefore use a dynamic difference-in-differences approach, which can accommodate recurrent and dynamic treatment.

The identifying assumption for this method is that the treatment and control groups exhibit parallel pre-treatment trends, conditional on treatment history, covariates, and the outcome variable. This assumption is satisfied in the data (Appendix 4).

In the subsequent analysis, treatment is a transaction involving a digital platform or digital ecosystem, while the unobserved potential outcome is the number of patents in the industry in the absence of such a transaction. A transaction may fail to occur for different reasons, including shocks on the side of the acquiring platform, industry-level shocks, or country-level shocks. In such a setting, inverse-probability

weighting could be an appropriate method. However, this type of estimation requires firm-level data, which I do not have because taxpayer identification numbers (INNs) are absent from the patent and M&A datasets. As a result, the two datasets cannot be merged without first aggregating them to the industry level.

Given this data structure and treatment pattern, I use a TWFE model estimated by ordinary least squares to study the relationship between the number of economic concentration transactions and the number of patents. To estimate the effect of economic concentration transactions involving digital platforms on innovative activity, I use a dynamic difference-in-differences model combined with propensity score matching.

3.2.2 TWFE Model Specification

I first estimate the relationship between the number of M&A transactions involving digital platforms in an industry and the number of patents produced in that industry. Following [Stavniychuk A.U., Markova O.A., 2023], I estimate a TWFE model with month and industry fixed effects to account for time shocks and industry-specific shocks. I estimate the following regression:

$$\begin{aligned} \log(Y_{it} + 1) = & \beta_1 \text{cum_platform_deals}_{it} + \beta_2 \text{capacity}_{it} + \beta_3 \text{cum_deals}_{it} \\ & + \beta_4 \text{cum_price}_{it} + \beta_5 \text{avg_patents_age}_{it} + \beta_6 \text{horizontal}_{it} \\ & + \beta_7 \text{lag_cum_patents}_{it} + \alpha_i + \tau_t + \varepsilon_{it}. \end{aligned} \quad (1)$$

where

1. Y_{it} is the number of patents in industry i in month t ;
2. $\text{cum_platform_deals}_{it}$ is the cumulative number of all transactions involving digital platforms in industry i by month t ;
3. capacity_{it} is a proxy for industry size, measured as the number of firms in industry i in month t ;
4. cum_deals_{it} is the cumulative number of transactions in industry i by month t ;
5. cum_price_{it} is the cumulative acquisition value in industry i by month t ;
6. $\text{avg_patents_age}_{it}$ is the average age of patents in industry i in month t ;
7. horizontal_{it} is the share of horizontal transactions in industry i in month t ;
8. $\text{lag_cum_patents}_{it}$ is the lagged cumulative number of patents in industry i in month t ;
9. α_i and τ_t are industry and month fixed effects;
10. ε_{it} is the error term.

I then test the specification for nonlinear relationships with the number of transactions, transaction value, and average patent age. I add the squared number of transactions involving digital platforms, squared average patent age, squared industry size, and the logarithm of transaction value to the specification (Model 2, Table 4).

I also test for a dynamic relationship by adding up to six lags of the number of transactions involving digital platforms (Models 3 and 4, Table 4).

In all specifications, I use the logarithm of the dependent variable because the density of its distribution has a logarithmic shape, while the dependent variable itself is nonnegative (Appendix 3). I also test whether the distribution of the number of patents resembles a Poisson distribution (Appendix 3). The results indicate that the distribution of the dependent variable does not correspond to a Poisson distribution.

3.2.3 Dynamic Difference-in-Differences Model

To estimate the effect of economic concentration transactions involving digital platforms on innovative activity, I use the Dynamic Difference-in-Differences method of [Imai, K., Kim, I.S., Wang, E., 2023]. The method first matches observations on their treatment histories, with the length of the treatment history chosen by the researcher. I use a four-month treatment history because of the bias–variance trade-off: the longer the treatment history, the lower the variance, but at the same time the number of observations that can serve as matches for a treated observation declines. The procedure then matches or weights observations on covariates. I test several approaches, including propensity score matching and propensity score weighting. Because the different methods provide a comparable degree of covariate balance, I use propensity score matching in the main specification.

This approach makes it possible to address the overlap problem by excluding observations with extreme propensity scores for which reliable matches cannot be constructed. At the same time, the region of common support remains sufficiently large to retain a representative subsample and support the subsequent estimation of treatment effects. I match on industry and date, a proxy for industry size, average patent age in the industry, the cumulative number of patents in the industry, the number of transactions in the industry, and the number of transactions involving digital platforms in the industry. I also test an alternative matching specification that separately accounts for the number of transactions involving digital ecosystems and the number of transactions involving all other digital platforms. I then estimate the effect of economic concentration transactions involving digital platforms on innovative activity over the twelve months following a transaction (Figure 12).

It is important to note that services and trade and telecommunications are unusual in that they have the largest number of transactions involving digital platforms, often in every month. I therefore test the

robustness of the results by excluding these industries (Figure 8). Moreover, because transactions occur continuously in these industries, the absence of an estimated effect may reflect habituation and a decline in sensitivity to repeated treatment.

I also conduct a placebo test that estimates the change in the outcome at time $t - 1$ relative to other pre-treatment periods within the lag window. These effects are statistically insignificant, supporting the robustness of the estimates (Appendix 2). In addition, I construct a placebo dataset in which the treatment variable is generated as follows. For each industry, I calculate the number of months in which transactions involving digital platforms actually occurred. I then randomly assign transaction dates to industries while preserving the actual number of treated months in each industry. I re-estimate all preceding specifications using this placebo dataset. The resulting estimates are statistically insignificant (Appendix 2).

3.3 Results

3.3.1 Relationship Between the Number of Digital-Platform Transactions and Patent Output (TWFE)

The TWFE estimates of the relationship between the number of transactions involving digital platforms and the number of patents produced indicate a negative association. A one-unit increase in the cumulative number of transactions involving digital platforms is associated with an approximately 4.3% decline in the number of patents on average, holding other factors constant (Model 4). At the same time, the cumulative number of all M&A transactions is also negatively associated with the number of patents in an industry (Model 3), suggesting that the level of competition in an industry is positively correlated with innovative activity. A 1% increase in cumulative transaction value is associated with an approximately 0.409% increase in the number of patents in an industry on average, holding other factors constant (Model 4). Average patent age in an industry is positively associated with the number of patents produced (Model 1).

I also estimate specifications with nonlinear relationships and find a U-shaped relationship between average patent age in an industry and the number of patents produced (the relationship takes the form of a downward-opening parabola). The logarithm of patent output increases until average patent age in the industry reaches 1,429 days, or approximately four years, and declines thereafter. The maximum average patent age observed in an industry-month is 2,205 days (Table 3). As average patent age rises toward 1,429 days, an industry may approach a saturation point. This may mean that by this stage most promising and highly profitable ideas have already been patented and fewer substantial new innovations remain. Firms may therefore begin to invest less in patenting because the expected return to new patents falls, especially when older patents protect technologies that have become obsolete or less relevant.

According to the estimated coefficients, there is no relationship between up to six lags of treatment and the number of patents produced (Table 4). As a robustness check, I also estimate models without fixed effects and find a robust negative association between the number of transactions involving digital platforms and the number of patents in an industry (Appendix 1). These specifications also exhibit a U-shaped relationship between average patent age in an industry and the number of patents produced.

Source: constructed by the author.

The result may be driven by external shocks, such as industry-level or month-level shocks, which can bias estimates from models without fixed effects. I therefore rely on the TWFE estimates when interpreting the results. Although the TWFE results are robust, some variables are measured only approximately; for example, industry size is limited to the number of firms observed in transactions. These results should not be interpreted as causal estimates. Under recurrent and dynamic treatment, classical approaches, including two-way fixed-effects regressions, can produce biased estimates. I therefore next estimate a Dynamic DiD model [Imai, K., Kim, I.S., Wang, E., 2023] to assess the effect of economic concentration transactions involving digital platforms on innovative activity.

3.3.2 Estimated Effects of Digital-Platform Transactions on Innovative Activity

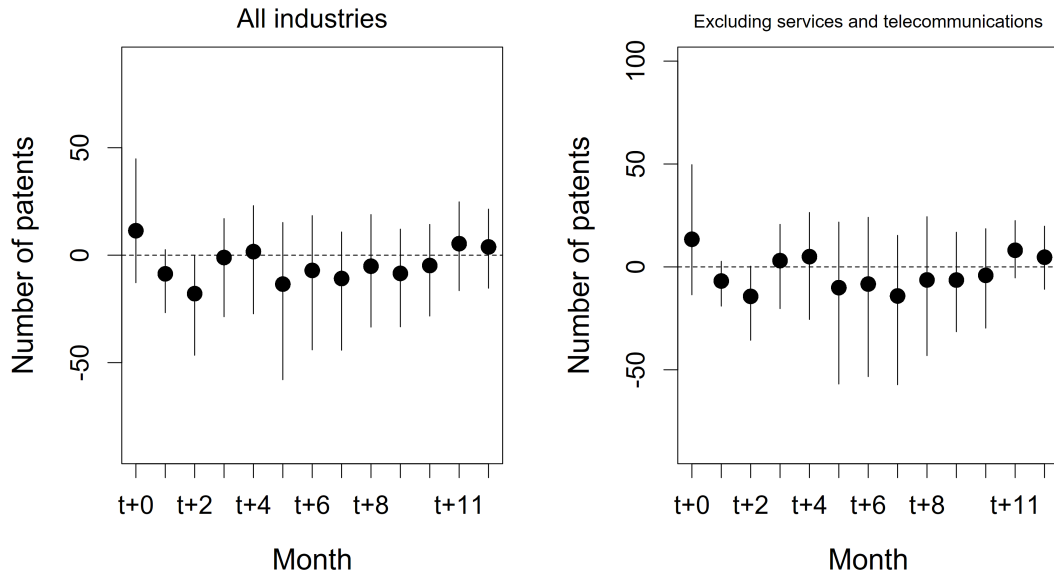
The Dynamic DiD estimates show that transactions involving digital platforms have a negative effect on the innovative activity of merging firms two months after the transaction. In the second month after the month in which a digital-platform transaction occurs in an industry, the number of patents in that industry falls by 19 patents (13.7%). This effect is temporary, however, and becomes statistically insignificant after three months. When services and trade and telecommunications are excluded, the negative effect disappears (Figure 12). The decline in innovative activity therefore appears to be concentrated in services and trade and in telecommunications, the industries in which transactions involving digital platforms—especially digital ecosystems—occur most frequently.

Figure 12. Effect of transactions involving digital platforms on innovative activity. Dynamic DiD estimates.

Table 4: Patent Output and Digital-Platform Transactions

	Dependent variable: $\log(\text{Patents} + 1)$			
	(1)	(2)	(3)	(4)
Cumulative platform transactions	0.0189** (0.0062)	0.0070 (0.0103)		
Average patent age	0.0010*** (0.0002)	0.0029*** (0.0003)	0.0010** (0.0003)	0.0030*** (0.0005)
Cumulative M&A transactions	-0.0019* (0.0007)		-0.0024* (0.0009)	
Cumulative transaction value	5.34×10^{-12} (1.00×10^{-6})		5.31×10^{-12} (1.00×10^{-6})	
Industry size	-0.0119** (0.0039)	0.0044 (0.0076)	-0.0086 [†] (0.0043)	0.0126 (0.0087)
Share of horizontal transactions	0.0252 (0.0256)	-0.0129 (0.0289)	0.0183 (0.0218)	-0.0381 (0.0316)
Lagged cumulative patents	-2.44×10^{-6} (6.39×10^{-6})	-1.10×10^{-5} (1.13×10^{-5})	-3.04×10^{-6} (7.26×10^{-6})	-1.00×10^{-5} (1.13×10^{-5})
Cumulative platform transactions squared		-4.56×10^{-6} (0.0002)		-1.51×10^{-5} (0.0002)
Average patent age squared		-9.59×10^{-7} (1.02×10^{-6})		-1.05×10^{-6} (1.03×10^{-6})
Log cumulative M&A transactions		0.0073 (0.1054)		-0.3116 (0.2187)
Log cumulative transaction value		0.0228 (0.0162)		0.1481 (0.0842)
Industry size squared		-0.0005 (0.0004)		-0.0006 [†] (0.0003)
<i>Distributed lags of cumulative platform transactions</i>				
Current period (t)			-0.0249 (0.0213)	-0.0430* (0.0187)
Lag 1			0.0007 (0.0308)	0.0163 (0.0329)
Lag 2			0.0326 (0.0383)	0.0252 (0.0357)
Lag 3			0.0306 (0.0281)	0.0337 (0.0264)
Lag 4			-0.0170 (0.0309)	-0.0148 (0.0305)
Lag 5			-0.0321 (0.0391)	-0.0298 (0.0403)
Lag 6			0.0332 (0.0297)	0.0206 (0.0287)
Industry fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Observations	1,204	1,204	1,134	1,134
R^2	0.958	0.962	0.958	0.961
Within R^2	0.194	0.259	0.205	0.268

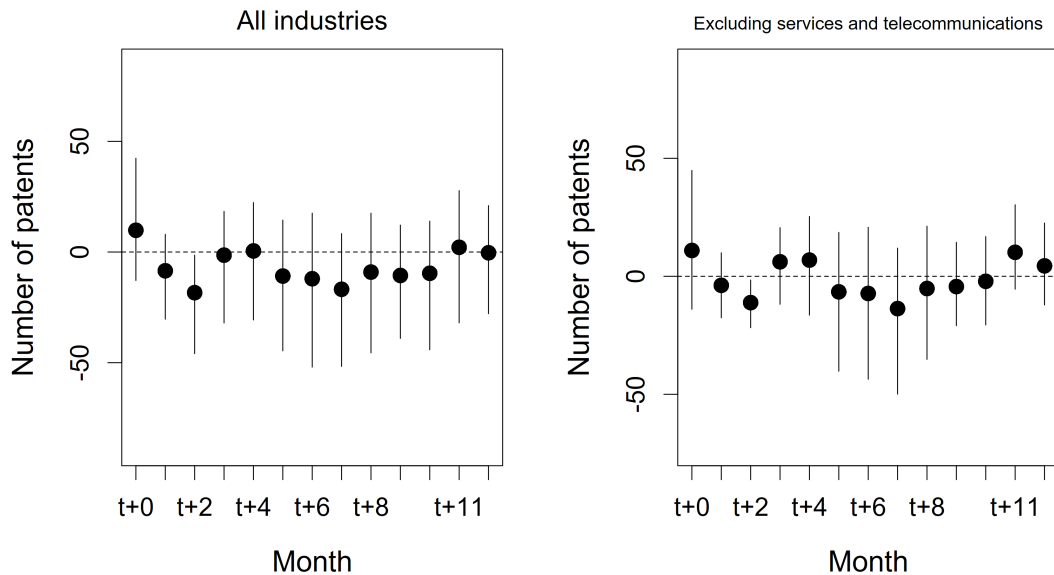
Notes: The table reports TWFE estimates of the relationship between digital-platform transactions and patent output. Standard errors, reported in parentheses, are two-way clustered by industry and month. Industry and month fixed effects are included in all specifications. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, and [†] $p < 0.10$.



Source: constructed by the author.

I next test an alternative matching procedure. Instead of matching control observations on the cumulative number of all transactions involving digital platforms in an industry, I separately match on the cumulative number of transactions involving digital platforms other than digital ecosystems and the cumulative number of transactions involving digital ecosystems (Figure 13). The estimates are robust: in the second month after a digital-platform transaction, the number of patents in the industry falls by 19 patents (13.7%).

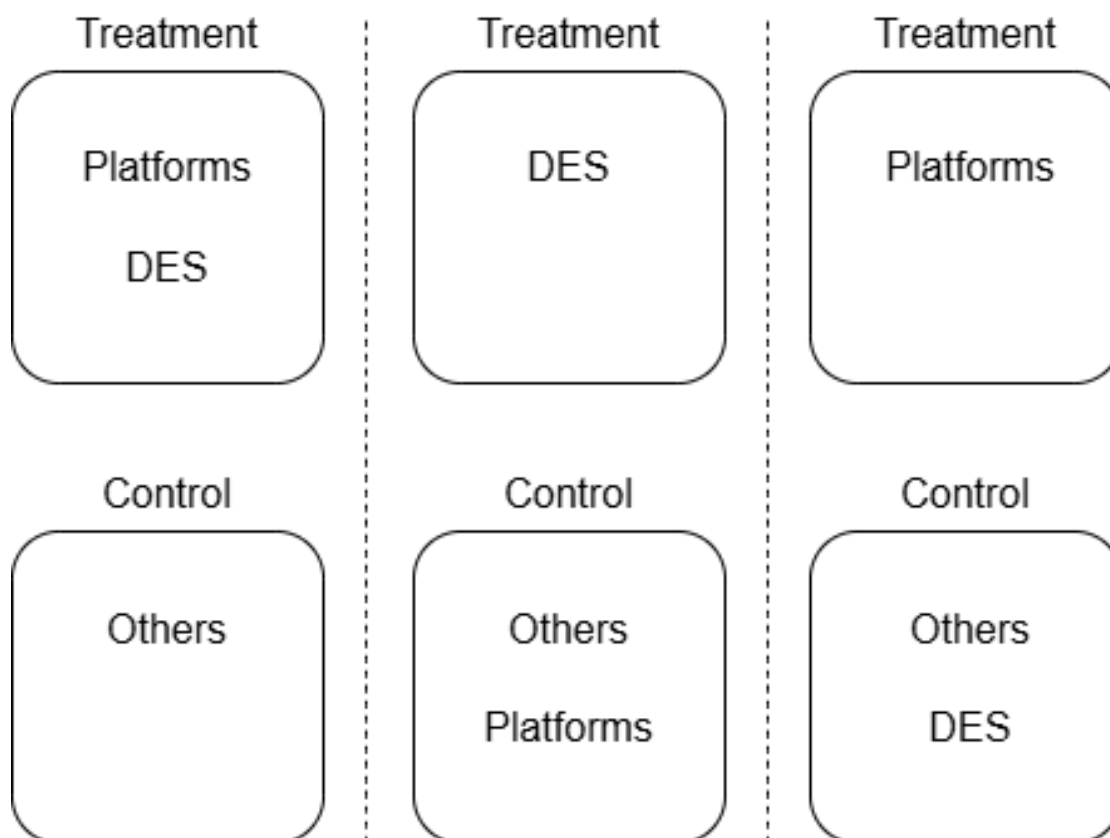
Figure 13. Effect of transactions involving digital platforms on innovative activity. Dynamic DiD estimates with alternative matching.



Source: constructed by the author.

This effect may reflect a decline in patent output in industries where a transaction occurs relative to industries where no transaction occurs. At the same time, the estimate may combine different mechanisms because digital ecosystems and other digital platforms are pooled together: the control group contains only industries in which, conditional on the matching variables, neither type of transaction occurred. The mutual influence of platforms and ecosystems is therefore embedded in the estimated effect. I consequently separate industries with transactions involving digital ecosystems from industries with transactions involving other digital platforms (Figure 14). If the estimated effects are similar, this would indicate that there are no substantial differences between the two types of platforms.

Figure 14. Treatment and control groups used to estimate the effects of transactions involving digital ecosystems and other digital platforms.

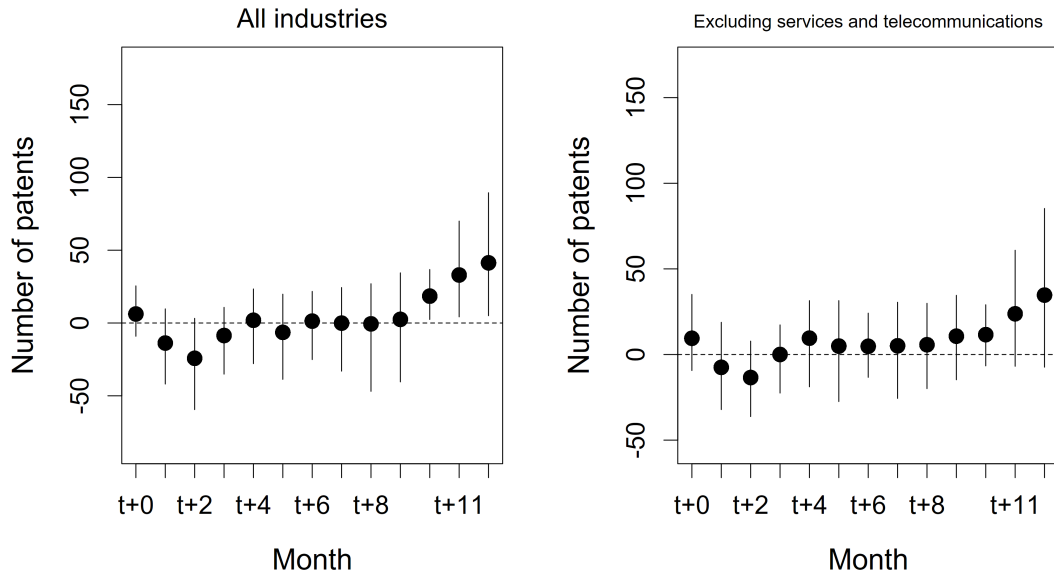


Source: constructed by the author.

Transactions involving digital ecosystems, by contrast, have a positive effect on the innovative activity of merging firms eleven months after the transaction. Eleven months after an economic concentration transaction involving a digital ecosystem, the number of patents in the industry increases by 30 patents (21.6%); after twelve months, it increases by 38 patents (27.3%). When all industries except services and trade and telecommunications are considered, the effect of digital-ecosystem transactions on innovative activity disappears, suggesting that the positive effect is concentrated in services and trade and in telecommunications (Figure 15).

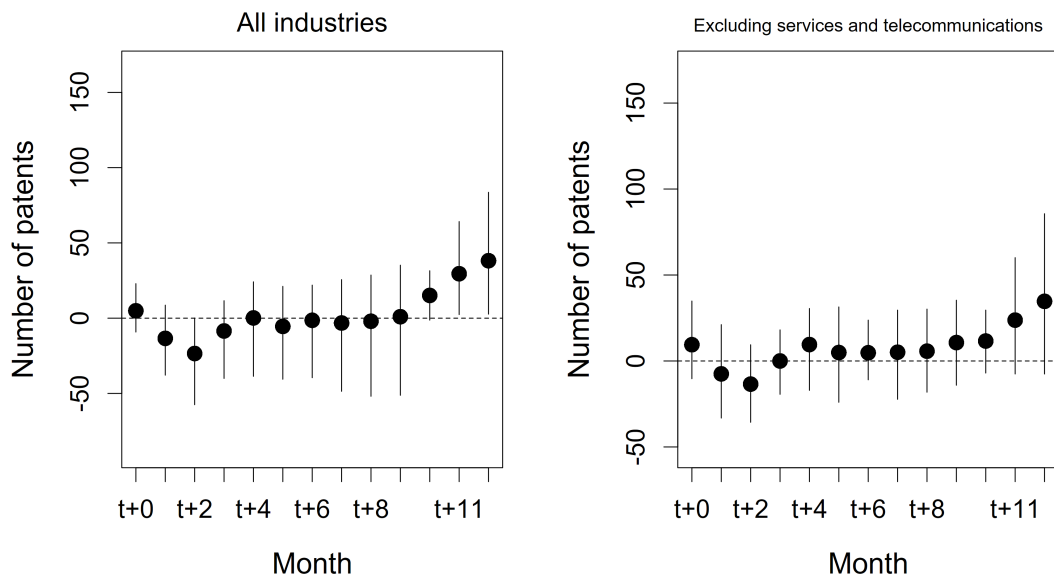
I also estimate a model that matches separately on the cumulative number of digital-ecosystem transactions and the cumulative number of transactions involving digital platforms other than digital ecosystems, rather than on the cumulative number of all digital-platform transactions in the industry. Eleven months after an economic concentration transaction involving a digital ecosystem, the number of patents in the industry increases by 30 patents (21.6%); after twelve months, it increases by 38 patents (27.3%). The estimates are robust to alternative matching procedures (Figure 16). Likewise, there is no effect of digital-ecosystem transactions outside services and trade and telecommunications. Under this specification, the control group includes not only industries with no digital-platform transactions but also industries in which transactions involving digital platforms other than digital ecosystems may have occurred.

Figure 15. Effect of transactions involving digital ecosystems on innovative activity. Dynamic DiD estimates.



Source: constructed by the author.

Figure 16. Effect of transactions involving digital ecosystems on innovative activity. Dynamic DiD estimates with alternative matching.



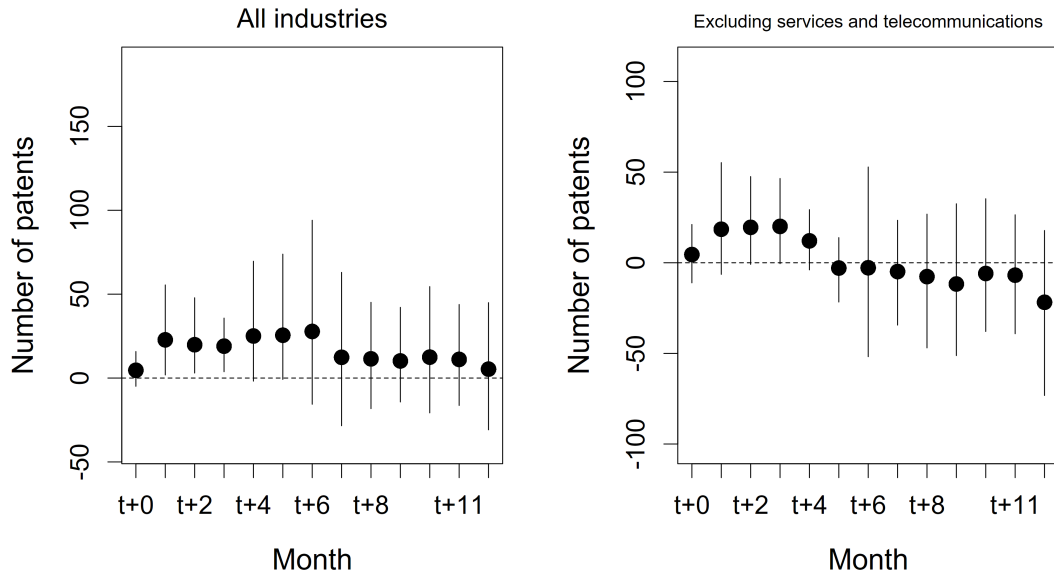
Source: constructed by the author.

In this case, it is possible that the observed effect does not arise from the production of new patents in the industry itself. Instead, when a digital ecosystem acquires a patent-holding company, other digital platforms in the same industry may respond by acquiring firms or producing patents of their own. One possible explanation is that digital platforms may be concerned that the digital ecosystem will make use of an antitrust exemption that it may obtain through newly acquired patents. This mechanism requires additional testing. For example, the analysis could be replicated around the removal of such exemptions; a change in the estimated effect would provide indirect evidence of the role of antitrust exemptions.

I next consider transactions involving all digital platforms other than digital ecosystems. These transactions also have a positive effect on the innovative activity of merging firms, similar to transactions involving digital ecosystems, but the positive effect emerges in the first month after the transaction. One month after an economic concentration transaction involving a digital platform other than a digital ecosystem, the number of patents in the industry increases by 24 patents (17.3%). After two months, it increases by 21 patents (15.1%), and after three months the effect declines to 17 patents (12.2%). When services and trade and telecommunications are excluded, there is no effect of transactions involving non-ecosystem digital

platforms on innovative activity, suggesting that the positive effect is concentrated in these industries (Figure 17).

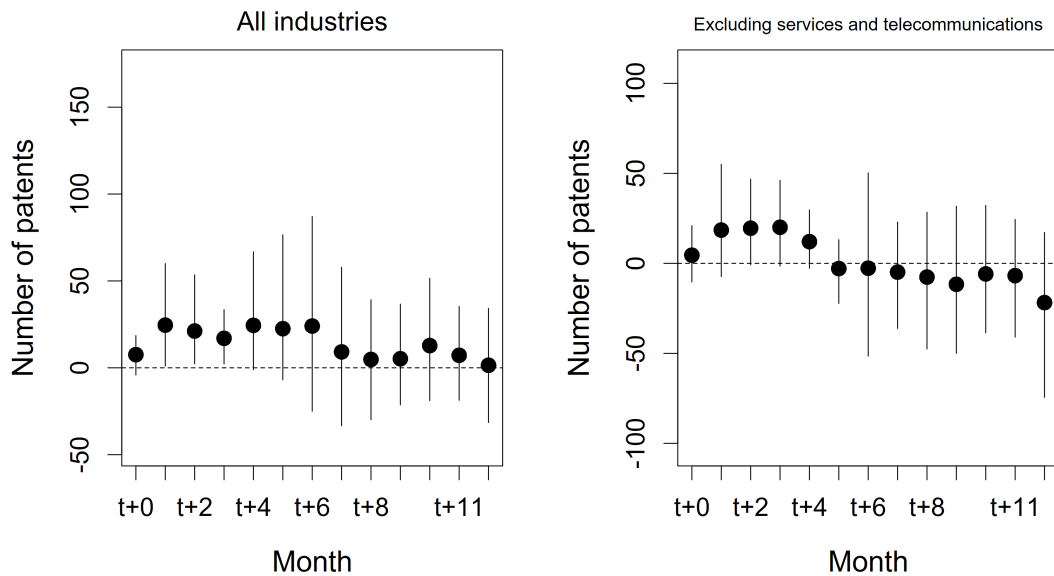
Figure 17. Effect of transactions involving digital platforms other than digital ecosystems on innovative activity. Dynamic DiD estimates.



Source: constructed by the author.

I also estimate the model using matching on the cumulative number of transactions involving digital ecosystems and the cumulative number of transactions involving digital platforms other than digital ecosystems, rather than on the cumulative number of all digital-platform transactions in the industry (Figure 18). One month after an economic concentration transaction involving a digital platform other than a digital ecosystem, the number of patents in the industry increases by 24 patents (17.3%). After two months, it increases by 21 patents (15.1%), and after three months the effect declines to 17 patents (12.2%). The estimates from the alternative matching specification coincide with the baseline results.

Figure 18. Effect of transactions involving digital platforms other than digital ecosystems on innovative activity. Dynamic DiD estimates with alternative matching.



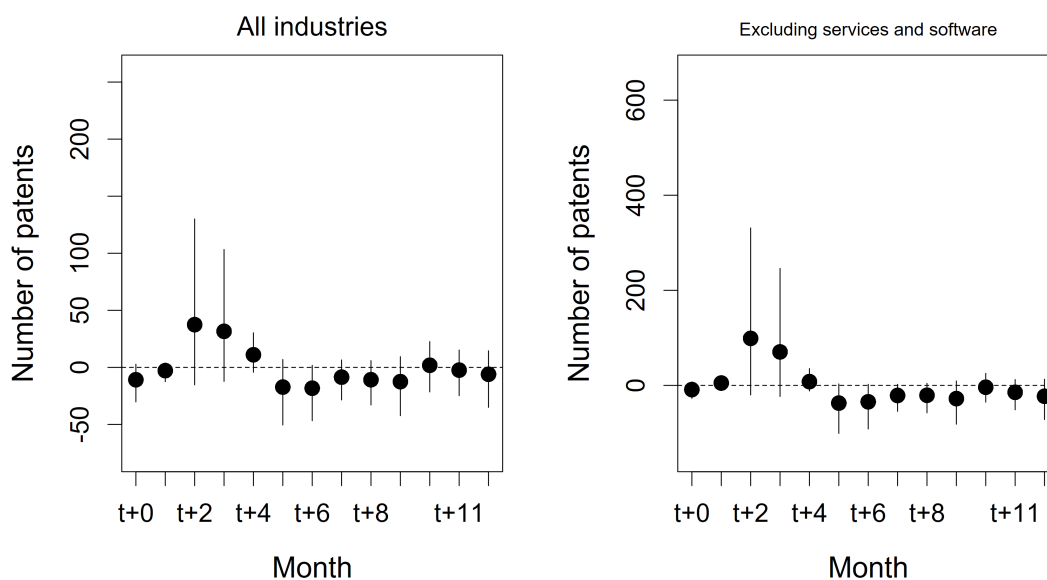
Source: constructed by the author.

Under this specification, the control group contains not only industries without transactions involving digital platforms but also industries in which transactions involving digital ecosystems may have occurred. It is therefore possible that the observed effect does not arise from the production of new patents in the

industry itself. Instead, in response to a digital platform acquiring a patent-holding company, digital ecosystems in the same industry may begin acquiring firms or registering new patents. Digital ecosystems may be concerned that the acquiring platform will make use of an antitrust exemption associated with newly acquired patents. In this case, the effect emerges in the first month after the transaction because digital ecosystems have greater resources for acquisitions and patent production than other digital platforms.

I next consider economic concentration transactions between digital platforms and startups. In this case, all coefficient estimates are statistically insignificant: I find no effect of digital-platform acquisitions of startups on innovative activity (Figure 19). No effect is detected when services and software are excluded from the analysis either.

Figure 19. Effect of digital-platform transactions with startups on innovative activity. Dynamic DiD estimates.

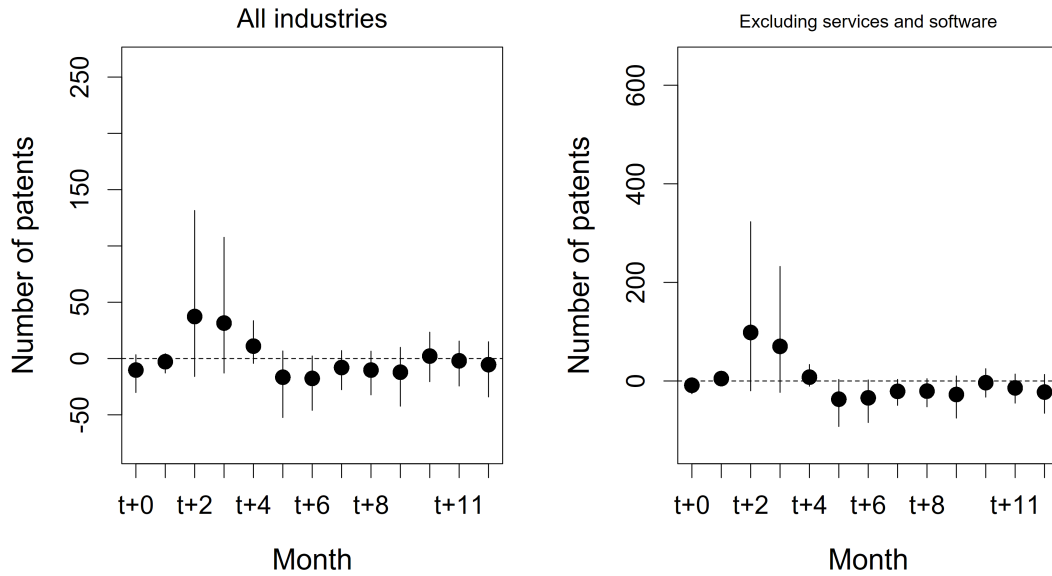


Source: constructed by the author.

Finally, I consider economic concentration transactions between digital ecosystems and startups. Here, too, I find no effect of digital-ecosystem transactions on innovative activity, and the result is robust to excluding services and software from the analysis (Figure 20). For startup transactions, the estimates for all digital platforms are nearly identical to those for digital ecosystems because almost all startup transactions were completed by digital ecosystems; only five transactions involved digital platforms that were not ecosystems. Startup transactions occur less frequently than transactions in the full sample (Figure 11). Moreover, the Rusbase data contain 34 industries, which is 20 more than in the Mergers data. The data are therefore less aggregated, and the absence of an effect may reflect this data structure.

The result can also be interpreted as evidence that acquisitions of startups by digital platforms do not affect innovative activity. This interpretation is consistent with the idea that, in the presence of network effects, selling to an incumbent may be an attractive outcome for startups. New startups may then enter industries in which acquisitions occur, while the total number of patents remains unchanged. In this case, any decline in patenting associated with the acquisition of a startup is offset by the entry of new startups.

Figure 20. Effect of digital-ecosystem transactions with startups on innovative activity. Dynamic DiD estimates.



Source: constructed by the author.

Overall, the analysis supports the hypothesis that economic concentration transactions involving digital platforms have a negative effect on innovative activity. The hypothesis of a U-shaped relationship between the number of economic concentration transactions involving digital platforms and innovative activity is not supported. However, I find a U-shaped relationship between patent output and the average age of patents in an industry. Monthly patent output increases until the average age of patents in the industry reaches approximately four years and declines thereafter (Table 5).

Table 5: Summary of Model Results

Method	Treatment	Data	Results
TWFE	Mergers	Platforms and digital ecosystems	Negative correlation between patent output and digital-platform transactions; U-shaped relationship with average patent age.
Dynamic DiD	Mergers	Platforms and digital ecosystems	Negative effect two months after a transaction.
Dynamic DiD	Mergers	Other digital platforms	Positive effect one to three months after a transaction.
Dynamic DiD	Mergers	Digital ecosystems	Positive effect 11–12 months after a transaction.
Dynamic DiD	Rusbase	Platforms and digital ecosystems	No statistically significant effect.
Dynamic DiD	Rusbase	Digital ecosystems	No statistically significant effect.

Source: constructed by the author.

Conclusion

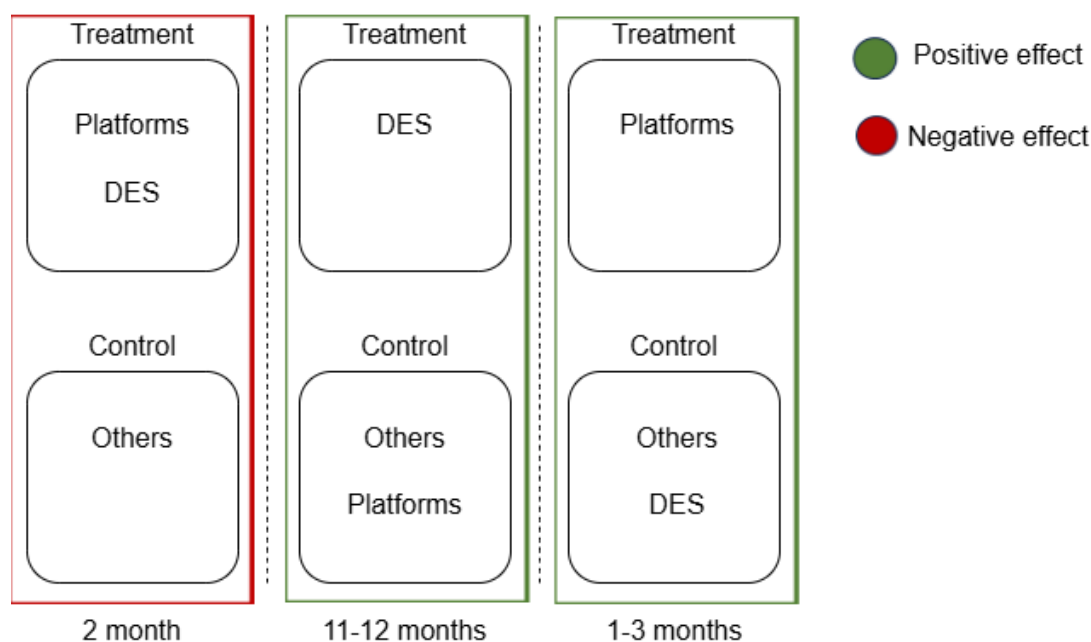
This study provides evidence of a negative effect of economic concentration transactions involving digital platforms on innovative activity. The hypothesis of a U-shaped relationship between the number of patents produced and the number of economic concentration transactions is not supported by the available data. I do, however, find a U-shaped relationship between the number of patents produced and average patent age in an industry. Monthly patent output rises until average patent age reaches approximately four years and then begins to decline. This may indicate that, by this point, most promising and highly profitable ideas have already been patented and fewer substantial new innovations remain. Firms may consequently begin to invest less in patenting because the expected return to new patents falls.

The hypothesis that economic concentration transactions involving digital platforms have a negative effect on innovative activity is supported by the estimates. Transactions involving digital platforms reduce the innovative activity of merging firms two months after the transaction. In the second month after the month in which an industry experiences a digital-platform transaction, the number of patents in that industry falls by 19 patents (13.7%). The effect is temporary, however, and becomes statistically insignificant after three months.

I also find two effects in the opposite direction: economic concentration transactions involving digital ecosystems and transactions involving digital platforms that are not digital ecosystems have a positive effect on the number of patents in an industry. These two effects should not be interpreted as an increase in innovative activity. One possible explanation is that the observed effect does not arise from the production of new patents by the directly affected firms. Instead, when a digital platform acquires a patent-holding company, digital ecosystems in the same industry may respond by acquiring other companies or registering new patents. Digital ecosystems may be concerned that the digital platform will make use of an antitrust exemption associated with newly acquired patents. In this case, the response appears in the first month after the transaction because digital ecosystems have greater resources for acquisitions and patent production than other digital platforms.

Similarly, digital platforms that are not ecosystems may respond to acquisitions by digital ecosystems. They may have incentives to acquire patent-holding companies in the same industry in order to avoid the consequences of an antitrust exemption held by a digital ecosystem (Figure 21).

Figure 21. Direction of the estimated effects of transactions involving digital ecosystems and other digital platforms.



Source: constructed by the author.

Future research could estimate the effect of changes in the antitrust regulation of economic concentration transactions involving digital platforms on innovative activity, examine the effect of digital-platform transactions on innovation quality—using patent citations as a possible proxy—and estimate firm-level effects, including heterogeneous effects. The main challenge for firm-level estimation is enriching the transaction and patent datasets with taxpayer identification numbers (INNs) for the firms involved.

Firm-level data would allow for more precise estimates, including controls for patent similarity—whether patents are substitutes or cover different technologies—and would make it possible to estimate the probability that a firm is acquired by a digital platform. This would allow researchers to analyze which firms digital platforms acquire, account for the size of the patent portfolio obtained through an acquisition, and estimate effects using inverse-probability weighting. Firm-level data would also make it possible to estimate the effect of changes in merger regulation using a triple-difference design.

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Appendix

Appendix 1. Estimates from Models without Fixed Effects

Table 8. Estimated Relationship Between Patent Output and Digital-Platform Transactions (No Fixed Effects)

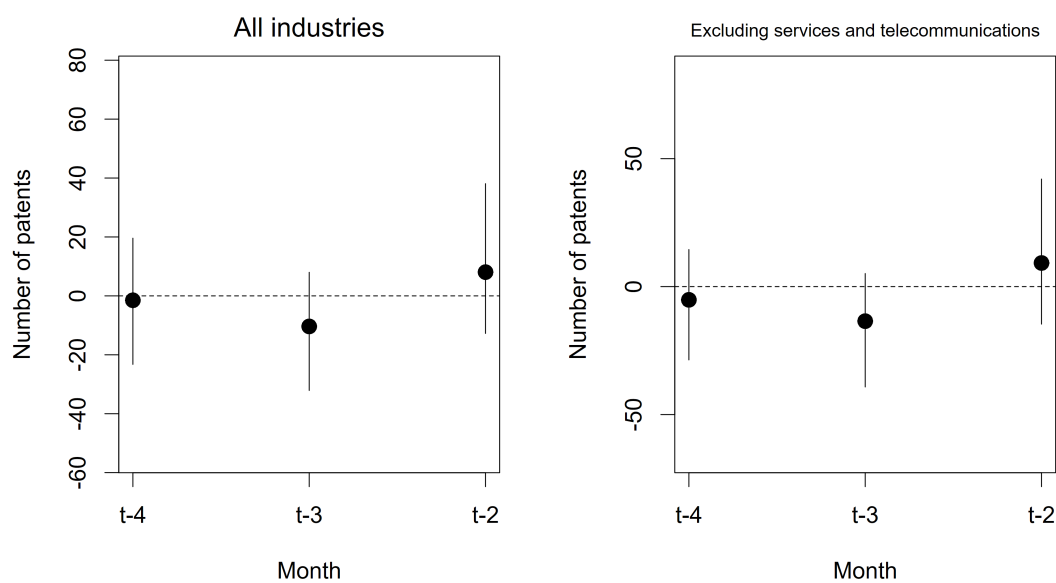
Dependent variable:	log_patents			
Model:	Model 1	Model 2	Model 3	Model 4
Constant	1.588*** (0.1041)	–1.390*** (0.4123)	1.560*** (0.1045)	–1.510* (0.6746)
Cumulative platform transactions	–0.0833*** (0.0060)		–0.1670* (0.0771)	
Average patent age	0.0018*** (7.44e-5)	0.0041*** (0.0002)	0.0020*** (7.92e-5)	0.0043*** (0.0002)
Cumulative M&A transactions	0.0100*** (0.0006)		0.0092*** (0.0007)	
Cumulative transaction value	–1.18e – 11 (1e-6)		–1.31e – 11 (1e-6)	
Industry size	–0.0033 (0.0109)	0.0287 (0.0281)	0.0168 (0.0142)	0.0074 (0.0317)
Share of horizontal transactions	0.1472 (0.1044)	0.1500 (0.1115)	0.1725 (0.1075)	0.1542 (0.1169)
log(cum_platform_deals+1)		–0.6740*** (0.0587)		–0.4215*** (0.1167)
Average patent age squared		–1.19e – 6 (1.01e-6)		–1.3e – 6 (1.01e-6)
Log cumulative M&A transactions		0.8073*** (0.0817)		0.8966*** (0.1008)
log(cum_price+1)		0.0169 (0.0274)		–0.0038 (0.0444)
Industry size squared		0.0009 (0.0014)		0.0031 (0.0018)
lag_cum_platform_deals_1			–0.0606 (0.1036)	–0.2281** (0.0780)
lag_cum_platform_deals_2			0.0176 (0.1010)	–0.0376 (0.1066)
lag_cum_platform_deals_3			0.0283 (0.1092)	0.0649 (0.1122)
lag_cum_platform_deals_4			–0.0112 (0.1184)	0.0240 (0.1164)
lag_cum_platform_deals_5			0.0020 (0.1203)	0.0018 (0.1159)
lag_cum_platform_deals_6			0.1250 (0.0816)	0.1266 (0.0770)
Cumulative platform transactions squared				0.0009*** (0.0002)
S.E. type	Heteroskedast.-rob.	Heteroskedast.-rob.	Heteroskedast.-rob.	Heteroskedast.-rob.
Observations	1,218	1,218	1,134	1,134
R ²	0.51227	0.53569	0.52884	0.55946
Adjusted R ²	0.50985	0.53262	0.52380	0.55355

Coefficient estimates with standard errors in parentheses. Significance codes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $p < 0.1$.

Source: constructed by the author.

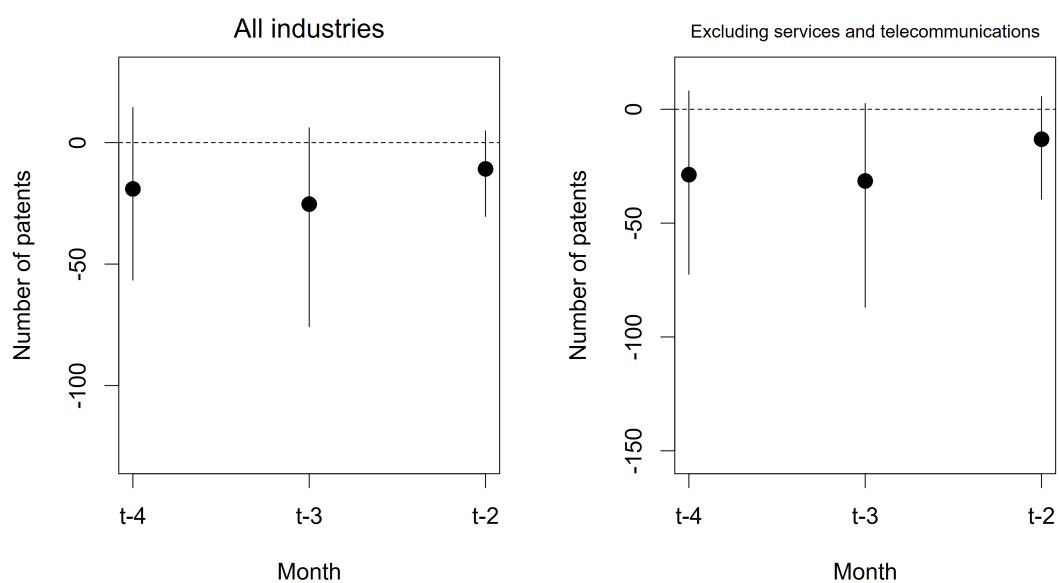
Appendix 2. Placebo-Test Results

Figure 22. Placebo test for digital-platform transactions in Mergers.



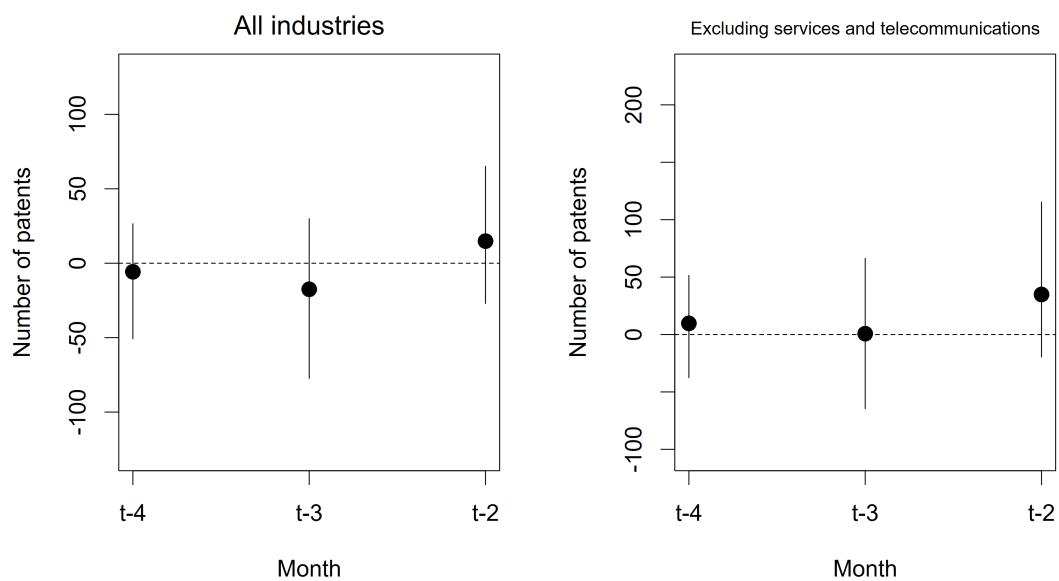
Source: constructed by the author.

Figure 23. Placebo test for digital-ecosystem transactions in Mergers.

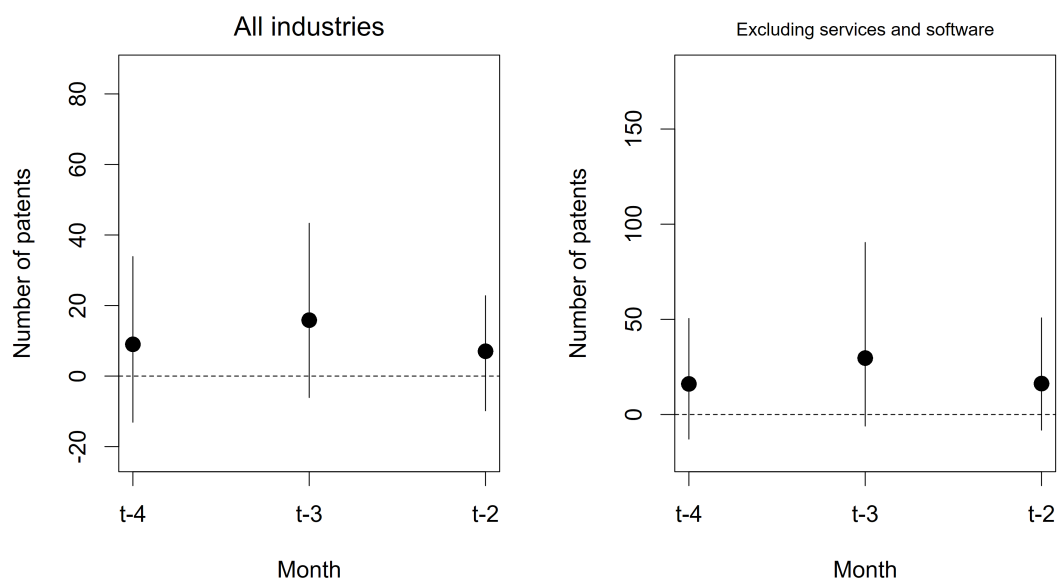


Source: constructed by the author.

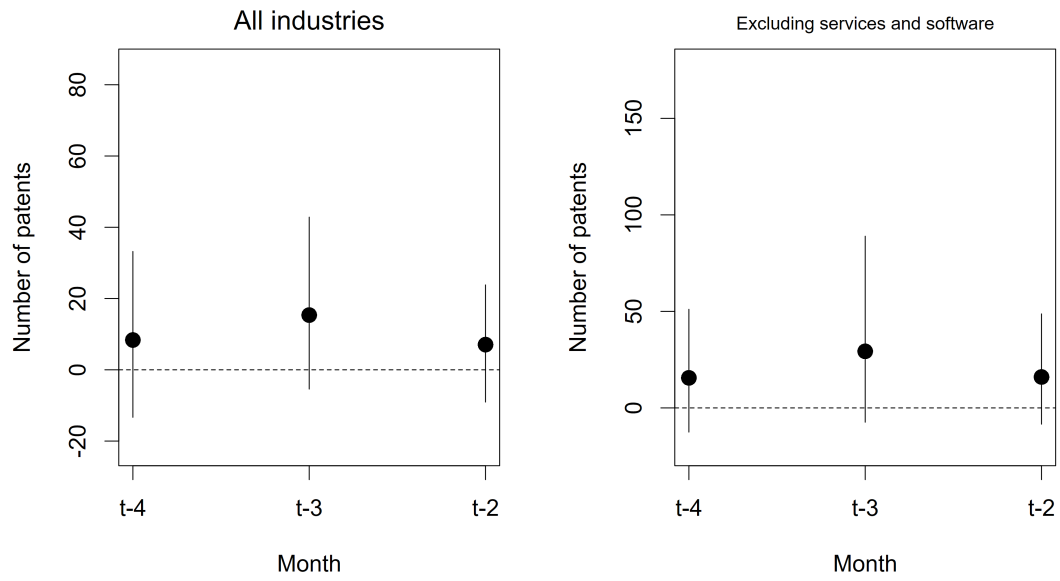
Figure 24. Placebo test for transactions involving digital platforms other than digital ecosystems in Mergers.



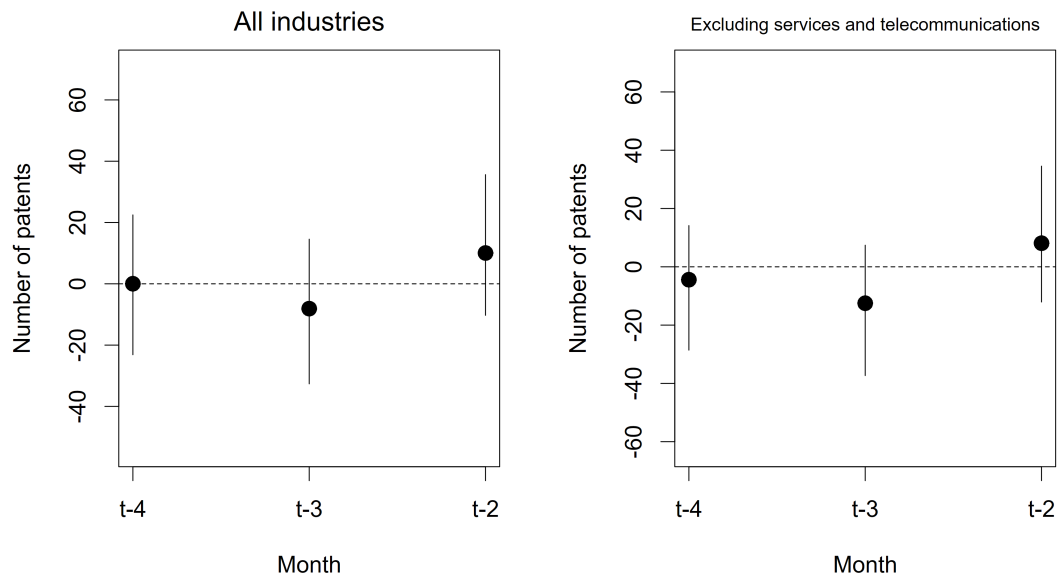
Source: constructed by the author.
Figure 25. Placebo test for digital-platform transactions in Rusbase.



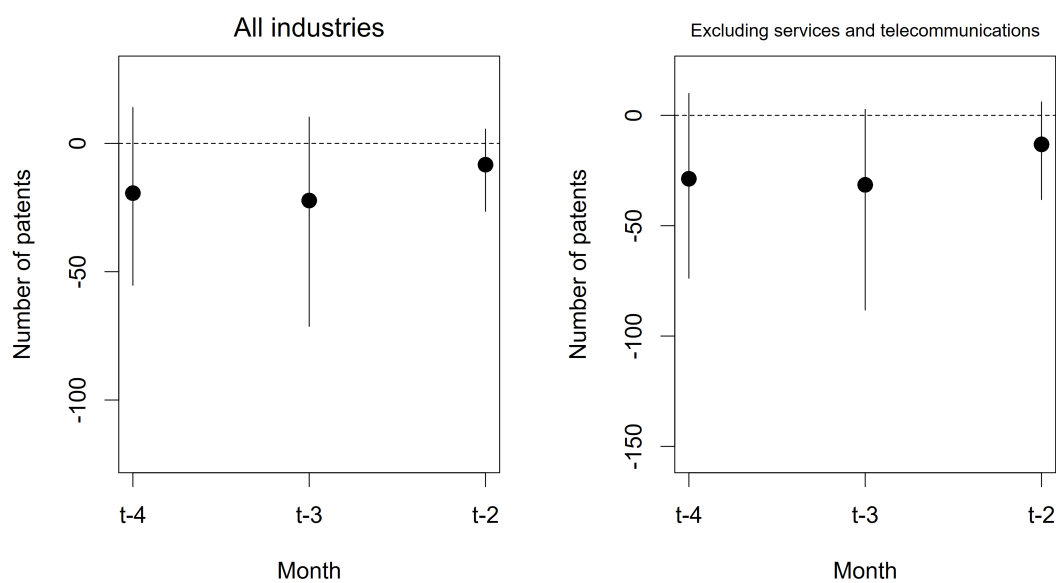
Source: constructed by the author.
Figure 26. Placebo test for digital-ecosystem transactions in Rusbase.



Source: constructed by the author.
Figure 27. Placebo test for digital-platform transactions in Mergers.

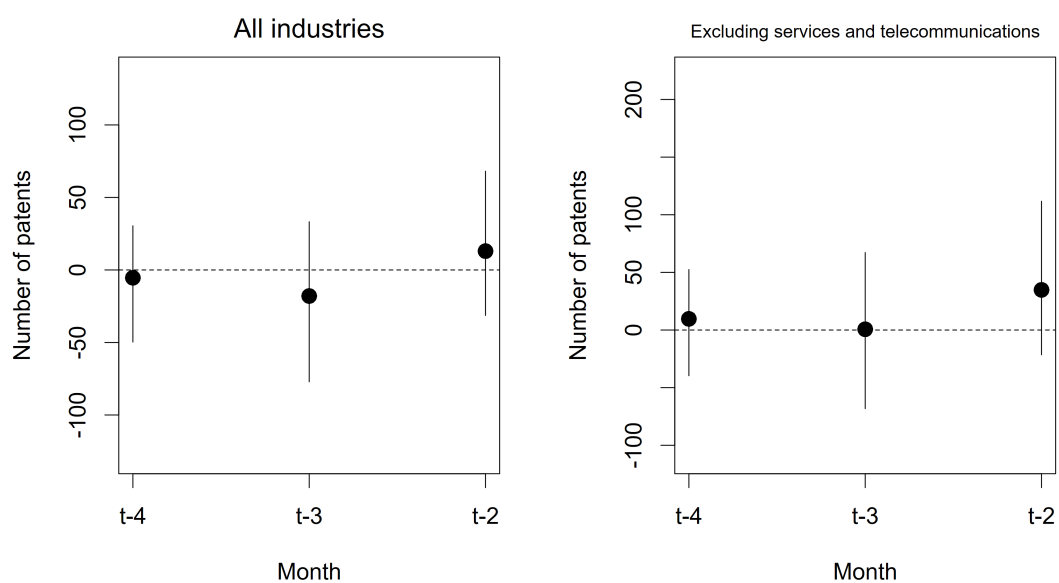


Source: constructed by the author.
Figure 28. Placebo test for digital-ecosystem transactions in Mergers.



Source: constructed by the author.

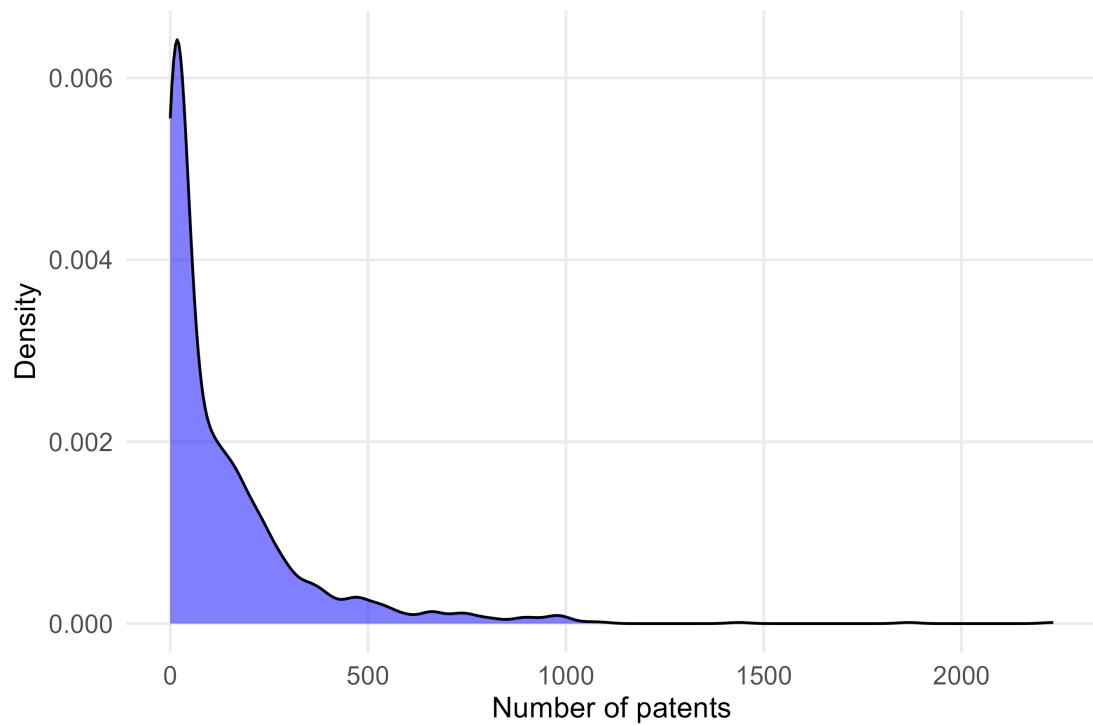
Figure 29. Placebo test for transactions involving digital platforms other than digital ecosystems in Mergers.



Source: constructed by the author.

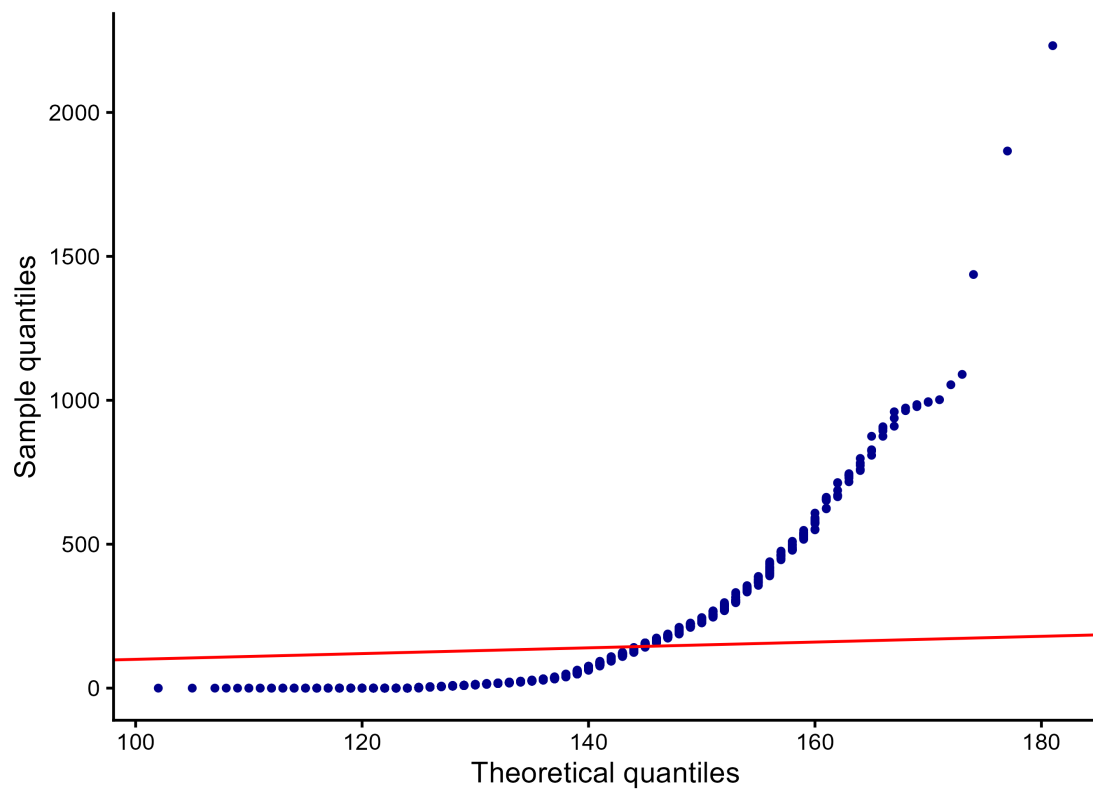
Appendix 3. Distribution of the Dependent Variable

Figure 30. Density of the patent-count distribution.



Source: constructed by the author.

Figure 31. Comparison of the dependent-variable distribution with a Poisson distribution.

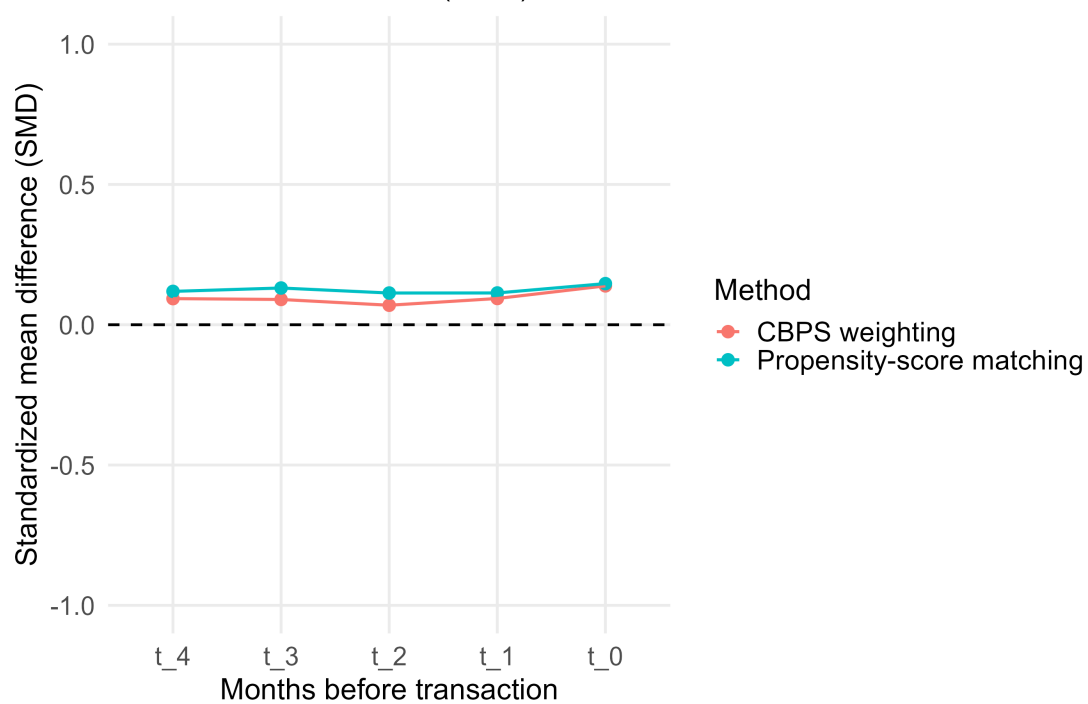


Source: constructed by the author.

Appendix 4. Identifying Assumption for the Dynamic Difference-in-Differences Method

Figure 32. Test of the parallel-pretrends assumption. Standardized mean difference (SMD) in patent counts between the treatment and control groups.

Standardized Mean Difference (SMD) in Patent Counts



Source: constructed by the author.